PROactive TEChnology-supported prevention and MEntal health in adolescence

PROTECt ME

Title: PROactive TEChnology-supported prevention and MEntal health in adolescence

1. General information

1.a Applicants

*Name**	*Institution*	*Department*	*Email*	*Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)*	*For Erasmus MC applicants: clinician/ researcher/ mix*	*Role in Flagship [Main Lead/Lead/ Applicant]*
Flagship lead						
Prof. Loes Keijsers	Erasmus University Rotterdam	Erasmus School of Social and Behavioural Sciences	keijsers@essb.eur.nl	Tenured		Main Lead
Prof. Manon Hillegers	Erasmus MC	Department of Child and Adolescent Psychiatry/Psychology (CAPP)	m.hillegers@erasmusmc.nl	Tenured	Mix	Lead
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Dr. Annabel Vreeker	Erasmus MC	Department of CAPP	a.vreeker@erasmusmc.nl	Other, temporary contract	Researcher	Applicant
Dr. Jeroen Legerstee	Erasmus MC	Department of CAPP	j.s.legerstee@erasmusmc.nl	Tenured	Mix	Applicant
Dr. ir. Ujwal Gadiraju	TU Delft	Faculty of Electrical Engineering,	u.k.gadiraju@tudelft.nl	Tenured		Applicant

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		Mathematics & Computer Science				
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Corresponding Lead: Prof. Loes Keijsers
Planned start date of seed funding: 31 Dec 2022
Primary Convergence theme: Technology Supported Transitions in Health(care).
Secondary Convergence theme (*if applicable*): Human Centered Technology and AI for Health

Keywords: *adolescents, mental health, technology, eHealth, Artificial Intelligence, prevention, real-time monitoring*

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Consortium partners (scholars who have expressed interest and support, who have helped drafting this proposal, and who will benefit from our infrastructure, meetings, insights and deliverables)						
Partners Erasmus University Rotterdam						
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Prof. Eveline Crone	Erasmus University Rotterdam	Erasmus School of Social and Behavioural Sciences	crone@essb.eur.nl	Tenured		Partner
Prof. Moniek Buijzen	Erasmus University Rotterdam	Erasmus School of Social and Behavioural Sciences	buijzen@essb.eur.nl	Tenured		Partner
Prof. Klaus Heine	Erasmus University Rotterdam	Jean Monnet Chair of Economic Analysis of European Law, Erasmus School of Law, Co-director Jean	heine@law.eur.nl	Tenured		Partner

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Partners Erasmus Medisch Centrum						
Prof. dr. Liesbeth van Rossum	Erasmus MC	Department of Internal Medicine, Division of Endocrinology, (internist-endocrinologist)	e.vanrossum@erasmusmc.nl	Tenured	mix	Partner

*Name**	*Institution*	*Department*	*Email*	*Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)*	*For Erasmus MC applicants: clinician/ researcher/ mix*	*Role in Flagship [Main Lead/Lead/ Applicant]*
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Dr. Mijke Lambregtse-van den Berg	Erasmus MC	Departments of Psychiatry, Child and Adolescent Psychiatry/Psychology, (child and adolescent) psychiatrist	mijke.vandenberg@erasmusmc.nl	Tenured	mix	Partner
Dr. Bram Dierckx	Erasmus MC	Department of Child and Adolescent Psychiatry/Psychology, child and adolescent psychiatrist, clinical pharmacologist	b.dierckx@erasmusmc.nl	Tenured	mix	Partner
Prof. dr. Neeltje van Haren	Erasmus MC	Department of Child Adolescent Psychiatry/Psychology	n.vanharen@erasmusmc.nl	Tenured	mix	Partner
Drs. Leontine ten Hoopen	Erasmus MC	Department of Child and Adolescent Psychiatry/ Psychology; Erasmus School of Health Policy & Management, Erasmus University Rotterdam, child and adolescent psychiatrist	l.tenhoopen@erasmusmc.nl	Tenured	mix	Partner

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Prof. dr. Pauline Jansen	Erasmus MC	Department of Child Adolescent Psychiatry/Psychology; Erasmus University ESSB	p.w.jansen@erasmusmc.nl	Tenured	researcher	Partner
Partners TU Delft						
Prof. dr. Mark Neerincx	TU Delft	Faculty of Electrical Engineering, Mathematics & Computer Science	M.A.Neerincx@tudelft.nl	Tenured		Partner
Dr. Frank Broz	TU Delft	Faculteit Elektrotechniek, Wiskunde en Informatica	f.broz@tudelft.nl	Tenured		Partner
Dr. Valentijn Visch	TU Delft	Faculty of Industrial Design Engineering	V.T.Visch@tudelft.nl	Tenured		Partner
Prof. dr. Pieter Jan Stappers	TU Delft	Faculty of Industrial Design Engineering	p.j.stappers@tudelft.nl	Tenured		Partner

Flagship societal partners in our broader community, collaborative partners and stakeholders (external parties who have expressed interest and support, who have helped drafting this proposal, and who will benefit from our infrastructure, meetings, insights and deliverables)

Name organization	*Description*	*Contactperson*	*Website*	*Contribution*
NVvP	The Dutch Psychiatric Association (NVvP) is a membership association of, for and by psychiatrists. More	Elnathan Prinsen, directiesecretariaat@nvvp.net	https://www.nvvp.net/home	**Letter of support.** Are willing to support this project with an active participation

	than 3,500 psychiatrists and psychiatrists in training are members of the association. The association aims to "promote psychiatry in the broadest sense, promote the scientific and professional interests of psychiatrists, promote and encourage education and training, and promote and monitor the quality of psychiatry.			of child and adolescent psychiatrists.
Municipality of Rotterdam	The municipality is responsible for youth mental health care.	Dr. Y.T.H.P. van Duijnhoven. director Public Health	https://www.rotterdam.nl/; https://www.rotterdam.nl/wonen-leven/zorgdichtbij/	**Letter of intent. For 200k euro's** In cash: 100.000 euros. In kind: 100.000 euros for recruitment support in youth Hubs, staff to organize workshops with youth, locations for knowledge exchange and youth participation.
PlusMinus	Dutch Patient Association for patients with bipolar disorder and their relatives	Henk Mathijssen, voorzitter@plusminus.nl	https://plusminus.nl/	**Letter of intent**. In kind: Support youth focus groups and facilitate dissemination of result. Representing a monetary value of **1040 euros.**
IJsfontein	IJsfontein designs and develops playing (digital) learning from	Hans Luyckx, info@ijsfontein.nl	https://www.ijsfontein.nl/	**Letter of intent.** In kind contribution: Building and

	the conviction that people are naturally curious and intrinsically motivated to develop themselves.			implementing machine learning in the GrowIt app. Representing a monetary value of **30.000 euros.**
Ethica Data	Ethica is the first platform that turns smartphones into micro research labs. It allows the use of almost all smartphone functionalities in research studies, from passive sensor monitoring to controlled tasks, and context-dependent surveys.	Mohammad Hashemian, mohammad@ethicadata.com	https://ethicadata.com/	**Letter of intent.** In kind contribution: **100.000 euros** to build data infrastructure
MIND and MIND Young	MIND wants to promote mental health and support all people involved. As an independent social organization we fight for a society that invests in psychological health and does everything to prevent unnecessary psychological suffering. We give a voice to all people with (starting) psychological complaints and their loved ones and offer information, advice and support - online, by telephone and anywhere in the country.	Drs. Marjan ter Avest, Director of MIND	https://wijzijnmind.nl	**Letter of intent**: In kind contribution: Support the recruitment of participants to test new mHealth tools, contribution to the feedback sessions in order to improve the mHealth application, use of the MIND Young platform to disseminate results and youth participation experiences. Representing a monetary value of **10.000 euros.**
Innovatic	Innovattic is a digital agency that creates digital solutions for societal challenges. They develop	Lauwerens Metz, lauwerens@innovattic.com	https://www.innovattic.com/nl	**Letter of intent**. In kind contribution: Consultancy on apps, module tracking teens.

	behavioral apps, impactful websites, serious games and more. Together with clients who, like us, believe in a better world.			Representing a monetary value of **25.000 euros**
VO Raad	The VO-raad is an association of schools in secondary education. The council represents the interests of secondary education with government, politics, business and social organizations. In addition, the Secondary Education Council promotes the quality of education by facilitating school administrators and school leaders in fulfilling their duties.	Rens van den Boogaard Teamleider School en Omgeving, VO-raad, info@vo-raad.nl	https://www.vo-raad.nl/over-de-vo-raad/verenigingsorganisatie/wat-doet-de-vo-raad	**Letter of support.** VO-raad intends to provide an in-kind contribution to support crosslinks between schools and researchers.This will include co-creation in research project, dissemination, supporting recruitment of mhealth study participating at all school levels (VMBO, MBO, HAVO, VWO), further detailed in the project proposal and budget form.
Jantje Beton	Jantje Beton strives for extra attention for children and young people in vulnerable (play) positions. They want to ensure that all children and young people in the Netherlands can play and exercise on a daily basis. Children have to do more and more and play less and less, while playing is the most important and the most fun part of a healthy life.	Pauline van der Loo, manager impact Jantje Beton, info@jantjebeton.nl	https://jantjebeton.nl/	**Letter of support.** Jantje Beton considers the active participation and empowerment of youth in research on this matter crucial for its success. Therefore, they support the cooperation between research and societal partners working with youth.

@ease	At @ease, all young people between the ages of 12 and 25 can spontaneously walk in or chat if they need someone to talk to: without a waiting list, free and anonymously. Trained young volunteers and professionals are on hand. Simply by being there and listening! And if necessary, we help find the right care outside of @ease.	Ruth Rietveld, Management @ease Rotterdam, ruth.rietveld@ease.nl	https://www.ease.nl/	**Letter of support.** @ease considers the active participation if youth and practitioners and empowerment of youth in research on this matter crucial for its success. Therefore, they support the cooperation between research and societal partners working with youth. They state that we intend to actively participate as advisors of the project as to share our knowledge, expertise, and networks.
Betasteunpunt Zuid-Holland	Bètasteunpunt Zuid-Holland connects secondary (secondary) and higher education (ho), a.o. through teacher professionalization and advancing inquiry-based learning and learning through (co-)design.	Renée Prins, Program manager Bètasteunpunt Zuid-Holland | Onderwijsnetwerk Zuid-Holland, r.j.prins-1@tudelft.nl	https://www.onderwijsnetwerkzuidholland.nl	**Letter of support.** advice, access to network in secondary education (SE) and support for organizing design projects on our themes in SE.
Wetenschapsknooppunt Delft	Science Exchange, Delft, researchers and designers work together with primary school teachers on projects in which the design process is central. The aim is to stimulate both students and teachers in creative thinking,	Remke Klapwijk, senior researcher at Wetenschapsknooppunt, R.m.klapwijk@tudelft.nl	https://www.tudelft.nl/wetenschapsknooppunt	**Letter of support.** advice, access to network in primary education (PE) and support for organizing design projects on our themes in PE.

	design and research.			
Healthy Start	Healthy Start is a new scientific initiative with the ambition to break through the unequal starting position in the lives of children and young adults.	prof. Eveline Crone, crone@essb.eur.nl	Convergence flagship, https://convergence.nl/nl/healthy-start/	Alliance between EUR, TUDelft and Erasmus MC on broader topic of child wellbeing. Provision of population data. Provision of network and experience regarding the use of EHealth. Access to student population, dissemination of results.
Generation R	Generation R is one of the largest cohort studies worldwide on adolescent mental health and investigates the growth and development of a new generation of Rotterdam children	prof. dr. Pauline Jansen & prof. dr. Manon Hillegers p.w.jansen@essb.eur.nl; m.hillegers@erasmusmc.nl	https://generationr.nl/	Provision of population data. Provision of network and experience regarding the use of EHealth. Access to student population, dissemination of results.
eHealthJr (including insurance companies, game designers, other academic partners;	Team of scientists, game designers, young people and companies, who took up this challenge to improve the quality of life of children and young people with a chronic disease using e-health	prof. dr. Manon Hillegers, m.hillegers@erasmusmc.nl	http://www.ehealthjunior.nl/	Provision of network and experience regarding the use of EHealth.
Student wellbeing program	The EUR Student Wellbeing programme aims to help students grow and thrive during their life here at the university.	dr. Danielle Remmerswaal, remmerswaal@essb.eur.nl	https://www.eur.nl/onderwijs/studeren-rotterdam/studentenwelzijn	Access to large network of Psychology students for dissemination of findings and help

				with data collections.
SYNC lab and Young Xperts	YoungXperts is an initiative of the SYNC lab, a research group at Erasmus University Rotterdam. We investigate what makes young people unique and how they find a role in society.	prof. dr. Eveline Crone, crone@essb.eur.nl	https://www.eur.nl/essb/onderzoek/erasmus-sync-lab; https://www.youngxperts.nl/over-ons/	Expertise on inclusion of adolescents.
Erasmus Centre for Data Analytics	Erasmus Centre for Data Analytics is to facilitate the Erasmus University and its public and private partner organizations in preparing society for a data-driven future, hands-on and human-centered, towards a sustainable world. To that end, the center houses, amongst others, one of the key Erasmus initiatives: Societal Implications of AI.	prof. dr. Moniek Buijzen, buijzen@essb.eur.nl	https://ecda.eur.nl/expert-practices/ai-dig-communication-behavioural-change/	Facilitation of data analytics.
RE-CONNECT	EUR/Erasmus MC	Prof. dr. Manon Hillegers; prof. dr. Pauline Jansen, prof. dr. Eveline Crone; prof. dr. Loes Keijsers. Contact: p.w.jansen@erasmusmc.nl	http://rotterdamexperts-connect.nl/	Alliance between municipality of Rotterdam and EUR/Erasmus MC

1.b Summary (298 words)

Societal problem. 13-25% of Dutch youth (aged 12-25) suffers from depression or anxiety. Such mental health problems have long-lasting negative effects, including educational dropout, worse physical health, and increased inequality. Mental health problems cost Dutch society 2 billion euro annually. Adequate and timely identification and treatment are currently impossible, due to long waiting lists, time-consuming monitoring procedures, and the lack of personalized, proactive prevention strategies. Hence, systematic changes in our approach to youth mental health are sorely needed.

Scientific ambition: Optimizing the well-being of Dutch youth aged 12 to 25 by:
(WP 1) improving early identification of mental health problems, using cutting edge real-time measures, smartphone and wearable technology, and data modeling;
(WP 2) improving decision making of who needs which intervention and when, combining professional expertise with artificial intelligence;
(WP 3) creating and evaluating new technology-driven proactive interventions (e.g., eHealth).

Our integrated approach involves stakeholders and end users in each step (WP 4); and builds a sustainable infrastructure and collaboration for convergence and impact (WP 5).

Output, outcome, impact. Apart from obtaining ground-breaking scientific insights into the antecedents of mental health problems, we will immediately provide new ways to improve detection, decision making and intervention, new methods and approaches, and user-friendly and cost-effective solutions for youth and professionals. In the long-term, this results in proactive, personalized, accessible, and cost-effective prevention of mental health problems, based on a sustainable business proposition. The ultimate impact will be a decrease in adolescents entering specialized mental health care, shorter waiting lists, a significant reduction in societal costs of adolescent mental health problems, and, most importantly, the optimal well-being of all youth.

Fit to convergence. Medical sciences, social science, humanities, and technology converge to promote this grand transition in mental health care, for those who need us most: Our children, our future.

1.c Public Summary (296 words)

Maatschappelijk probleem. 13-25% van de Nederlandse jongeren (12-25 jaar) ervaart depressieve gevoelens of angst. Dit heeft langdurige en ernstige negatieve effecten, zoals schooluitval, slechtere lichamelijke gezondheid en grotere kansenongelijkheid. Het leidt ook tot maatschappelijke kosten van wel 2 miljard euro per jaar. Behandeling komt vaak te laat en wachtlijsten zijn lang. In het huidige systeem is er weinig aandacht voor preventie en een systematische veranderingen van ons gezondheidssysteem is daarom hard nodig. Hiertoe, echter, ontbreken adequate kennis, monitoring-tools en toegankelijke interventies zoals eHealth apps.

Wetenschappelijke ambities: Het optimaliseren van welzijn van Nederlandse jongeren tussen de 12 en 25 door het:
Verbeteren van vroege identificatie van psychische problemen met geavanceerde real-time metingen en data-analyse;
Verbeteren van de besluitvorming over wie welke interventie wanneer nodig heeft door expertise van professionals te combineren met artificiële intelligentie; en
Ontwikkelen van nieuwe technologisch-gedreven interventies, bijvoorbeeld eHealth.

Bij elke stap betrekken we eindgebruikers (jongeren, professionals) en stakeholders (opvoeders, professionals, beleidsmakers) en we bouwen ook aan een duurzame infrastructuur en samenwerking tussen wetenschappers en de maatschappij.

Opbrengst. We ontrafelen de vroege kenmerken van geestelijke gezondheidsproblemen en creëren nieuwe manieren om emotionele problemen eerder en beter op te sporen en jongeren sneller te verwijzen. We bouwen aan een nieuwe generatie gebruiksvriendelijke en kosteneffectieve innovatieve tools voor jongeren en professionals, waaronder eHealth toepassingen. Zo dragen we bij aan veilige en toegankelijke tools ter preventie van emotionele problemen.

Impact. De meetbare maatschappelijke impact hiervan is: een afname van het aantal jongeren dat instroomt in de geestelijke gezondheidszorg (GGZ), kortere wachtlijsten, en lagere maatschappelijke kosten van mentale gezondheidsproblemen. Maar vooral, optimaal welzijn van alle jongeren.

Convergentie. Medische wetenschappen, sociale en geesteswetenschappen en technologie komen samen om deze grote transitie in de geestelijke gezondheidszorg te bevorderen, voor degenen die ons het meest nodig hebben: Onze kinderen, onze toekomst.

2. Relevance to Convergence Health & Technology mission (2444 words)

2.a Contribution to H&T mission and fit with H&T theme(s)

TRANSITION

Urgent societal problem. The mental health of adolescents (aged 12-25) is under serious threat: According to UNICEF, nine million European adolescents have mental disorders, most dominantly anxiety and depression [1,2]. Approximately 13-25% have mental health complaints [3]. The demands on Child and Adolescent Mental Health (CAMH) services are steadily increasing, and the societal and medical costs have exploded to a staggering 49 billion euro annually in Europe [2].

These costs also expand to families and society and carry forward to future generations. Mental health problems, even more so than medical conditions, may cause irreversible damage to a developing brain, seriously reduce long-term educational attainment, increase welfare programs dependency, worsen life satisfaction, and reduce physical health. People with severe mental illnesses die, on average, 20 years younger [4].

Current situation. Often emotional problems are not timely recognized. When they are, treatment is often delayed [5]. Even though access to necessary mental health services is a human right, Dutch adolescents wait on average 3-4 years before depression is adequately treated [6]. The youth mental health-care system is overburdened: Professionals are unable to handle the demand, waiting lists are long, and the system is no longer financially sustainable.

Desired situation. In 2030, each adolescent should have access to engaging preventive tools, tailored to his or her needs, which are fun to engage with and promote well-being before emotional problems emerge.

MISSION & AMBITION

Perfectly aligned with the **overarching mission of the convergence Health & Technology,** we aim for life-long health and well-being, from pre-birth to end of life in health(care) and for quality and equity in health care, through early diagnostics to proactively maintain health and individually tailored precise medical prevention and treatment for all.

The mission of this program is to optimize well-being of all youth, by proactively tackling these problems, to move from cure to care, and to detect emotional problems before they emerge and impair lives.

Ambition. Leading on from our extensive collaborations with stakeholders and our scientific analysis, the most cost-effective and promising solution to mental health problems is to transform our methods, research and health system from curing serious problems to proactive caring about youth and preventing problems. To achieve this outcome and impact (see also impact plan, paragraph 2.e) we will first of all fill the void of scientific insights into the causes of mental illness and then apply these insights to develop much needed, but still absent user-friendly and cost-effective methods and approaches to improve 1) identification and diagnosis, 2) decision making and 3) intervention for youth and professionals.

OBJECTIVES

We hypothesize that new technology and innovative methods will allow us to:
1) **identify and diagnose adolescents** at risk at a much earlier stage, supporting a better outcome;
2) **improve decision making** on who needs to be referred to which intervention, and;
3) **create and evaluate technologically driven interventions** to support youth in proactive, personalized, and timely manners.

TECHNOLOGICAL / SCIENTIFIC BREAKTHROUGHS

These major objectives are achieved by converging cutting-edge novel technology and methods from behavioral sciences, data-science and human centered design with emerging clinical needs and opportunities in practice. In doing so, we will take a world-leading role in transforming big data per person - to big solutions for society. This program, is built on the following technological/scientific breakthroughs (each addressed in a dedicated Work Package (WP, see Figure 3):

WP1) We will introduce a new toolbox with behavioral sciences methods to monitor real-time functioning of adolescents with smartphone apps – and create a center of methodological and clinical expertise around these so-called experience sampling methods [7-9]. This will allow us to identify whether a given individual is at risk in real-time, in real settings, leading to earlier diagnostics in WP1.
WP2) We will utilize the increasing possibilities of combining professional expertise, the power of AI-aided algorithms from data-science with novel methods for user-centered design in WP2 [10-14].
WP3) Merging existing clinical data and new types of real-time data with state-of-the-art AI techniques is unprecedented and can be directly translated to new solutions for clinical identification, self-monitoring and prevention in WP3.
WP4) We will deliver a novel toolbox for transformative research on depression and anxiety with adolescents from diverse backgrounds and with varying risk profiles (e.g., from low to high risk) and other stakeholders as co-researchers in WP4.

IMPACT

Breaking new scientific ground, this program will contribute to decreasing the number of youth entering mental health care, which will shorten waiting lists, and reduce societal costs of adolescent mental health problems (see impact plan, Paragraph 2.e). These moonshot ambitions align perfectly with the strategic themes of *Technology Supported Transitions in Health(care)* and *Human Centered Technology and AI for Health*.

2.b Synergy between disciplines

Merging approaches / insights. Our newly established and rapidly expanding consortium and broader community integrates the knowledge, tools, and ways of thinking of a well-organized team of both established and highly talented young and mid-career scientists. Specifically, we converge medical knowledge on mental health(care) (Erasmus Medical Center=Erasmus MC), knowledge on (micro-econometric analyses of) adolescent development and prevention and intervention programs (Erasmus University Rotterdam=EUR), and ethics, motivational design, and AI (TU Delft=TUD) (Figure 1).

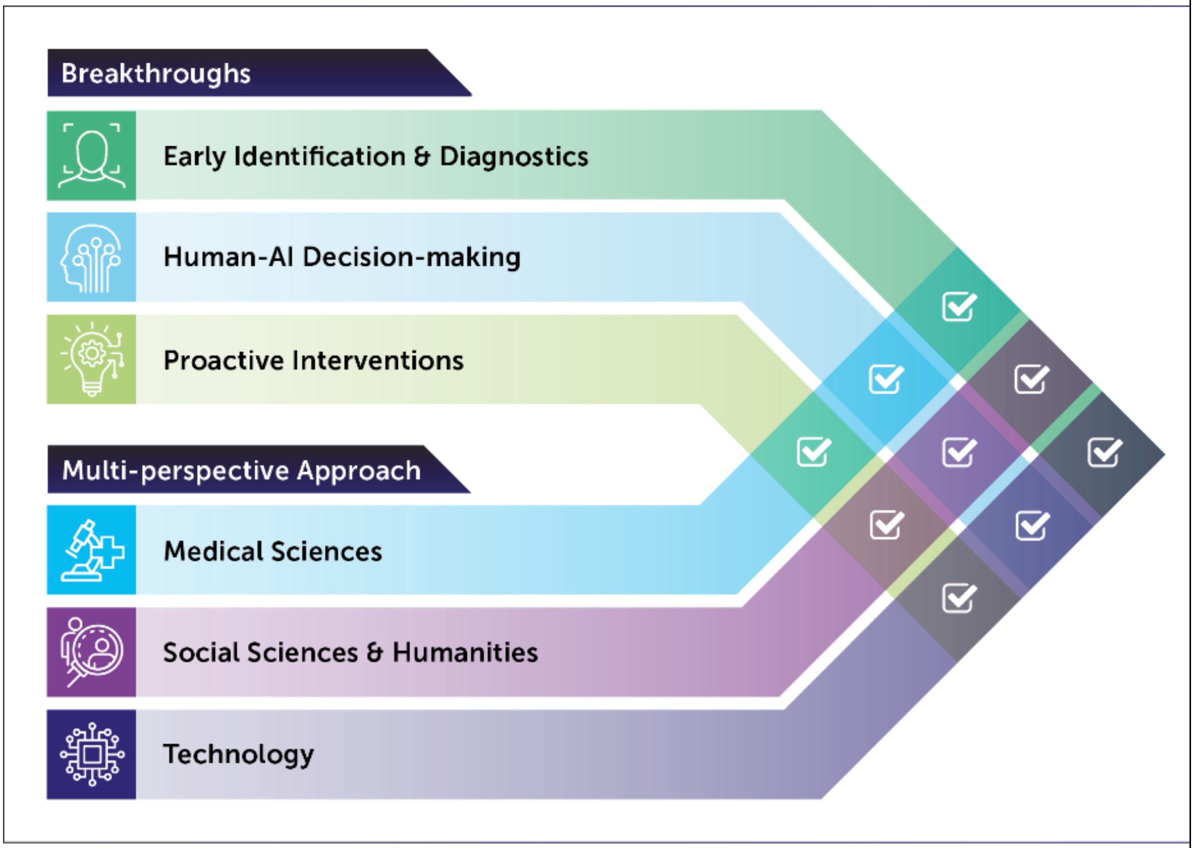

Figure 1. Merging scientific perspectives and approaches, to break new ground across disciplines.

Here we provide concrete examples, per work-package (WP), of how our goals and new scientific breakthroughs are reached through merging approaches and insights:

WP1. Identify and diagnose risk for emotional problems at a much earlier stage by combining cutting edge real-time measures and technology from behavioral sciences (EUR/Erasmus MC), such as Experience Sampling of daily lives' emotions and passive monitoring (EUR/TUD), with state-of-the-art data science (TUD) and statistics from social sciences (EUR) and epidemiology (Erasmus MC).

WP2. Improve decision making of who needs which intervention and when, by combining professional expertise from psychological (EUR) and psychiatric (Erasmus MC) practice with the power of Human-Centered AI techniques and algorithms from data-science (TUD). Moreover, user-centered design (TUD) is used to translate these algorithms and insights into effective

decision tools for professionals and adolescents (Erasmus MC/TUD, partners).
WP3. Create and evaluate technologically driven interventions to proactively support youth, by designing personalized and timely solutions together which meet the needs of children and professionals (EUR/TUD), such as gamified eHealth (Erasmus MC/EUR). We will also evaluate them on their (cost-)effectiveness (EUR), and ethical and legal aspects (EUR/TUD), in (clinical) practice (Erasmus MC).
WP4. The human central. Tackling these grand challenges will be successfully done by putting the human central, which converges state-of-the-art research, design, and practical methods and expertise of all three institutes (Erasmus MC/EUR/TUD). Involvement of stakeholders in each WP from the beginning will safe-guard attention for the human experience and the ethical and legal aspects of our innovations, increasing positive and sustainable impact.

SYNERGY

Together, converging all of our methods, approaches, and insights into designing, testing, and disseminating new data-driven digital and enabling health technologies will pave new routes towards an affordable health(care) system, where adolescent mental health is proactively identified and prevented. By generating personalized solutions[15], we contribute to a more inclusive approach to mental health, which is facilitated throughout our work plan by the participation of adolescents themselves.

2.c Long term sustainability plan

Long-term interdisciplinary collaborations enable a proactive stance: Instead of merely following up on each other's work, resulting in a long research pipeline, we address each problem and challenge it rapidly from multiple perspectives. To ensure long-term sustainability of the program after the 5-year central funding period, we will have two supporting WPs (Figure 3), as described below.

WP5. Sustainable Impact. To conduct excellent science, to grow as a consortium, to strengthen bonds with academic and societal partners, and to build a sustainable business proposition, WP5 builds a comprehensive framework of inter- and transdisciplinary collaboration. This includes **collaborative activities**, such as:

- Half-yearly symposia and colloquia with external keynotes;
- Joint supervision of all PhD- and postdoc projects from at least two institutes (and 3-5 additional advisors);
- Novel interdisciplinary educational modules (e.g., MOOCs, master programs).

WP5 also creates the needed **infrastructure and facilities** to adhere to all ethical principles and good research practices. Amongst others, we will:

- Improve our data-management strategy for scientific research in clinical practice of eHealthJr, which allows us to safely and confidentiality collect, store, use and share data for scientific research and personalized treatments;
- Train community members in the FAIR principles of data management, and in Open Science principles and practices (e.g., preregistration);

- Build a local data-collection infrastructure for ESM studies (PPS with software company Ethica Data), to ensure safe, secure, and freely accessible tools for scientific data-collection through smart-phones, for researchers in the three institutes.

WP6. A strong governance. WP6 monitors and steers the consortium in agile modes of cooperating, and reaching its own ambitions, viability, and growth**.** The program (as explained in Section 4.d.) is led by an experienced daily board. The board will monitor the cooperative workflow, detect emerging risks and opportunities and effectively and proactively deal with it (see Section 4).

2.d Visibility, attracting talents and new partners

Scientific & societal visibility. Apart from including stakeholders in each step of our program (WP4), the consortium will actively reach out, by organizing **activities**, disseminating findings to **general audiences**, and generating new routes towards impact for society. To generate and monitor translation of output to outcomes and create lasting impact of the program prof. van Buuren (impact lead EUR) will help us design and improve our **impact plan** (Figure 3) throughout the program. To establish and monitor academic impact, our **advisory board** will be composed of international experts, and national stakeholders to critically advise the program on the scientific direction and societal impact.

Education. Our students should become leading academics with excellent skills to conduct interdisciplinary research. Therefore, already in Year 1, the members of the consortium will co-supervise theses within existing **MSc/MA programs** and provide guest lectures at each other's institutes. In Year 2, we will establish an interdisciplinary **BSc/BA minor** which provides access in Year 3 of the program, to our **new interdisciplinary master's program**: *Designing mental health.* Students from all disciplines can enter and will collaborate with societal stakeholders on solving real world problems related to mental health of the young. These students do scientific internships at (at least) 2 of the institutes in our community.

Attracting Talents. Our community will connect exceptionally strong creative young talents (e.g., VENI,VIDI,ERC Laureates) to senior scientists (e.g., Spinoza winners) and PIs of interdisciplinary consortia (e.g., NWA-ORC; Gravitation) and large research projects (e.g., Generation R, Student well-being program). Being **an early career scholar** within our flagship program, will be a truly interdisciplinary experience. Each young scholar will be **supervised by scholars from two disciplines**. Our **talent management program** includes a young board (PhD-students, postdocs) that will organize additional activities (e.g., colloquia) and advises the daily board and **personal mentorship**. We will grow substantially by attracting external funding for hiring personnel. Section 4d describes our strategic plans for doing this.

Attracting partners for social impact. The consortium is already well-connected to commercial and societal parties such as Philips, the Municipality of Rotterdam, and insurance companies. This should give us a solid start to help us grow into sustainable business propositions and stay aligned with all the existing and new legal, medical device regulations, and ethical and data-privacy and security concerns. The faculties that we will build (e.g., new master modules, research infrastructure) and the tools and products that we will deliver (e.g., eHealth, decision tools) will make this flagship highly attractive for private-public collaborations. Erasmus Research Services will be our advisor in this process.

2.e Long term societal impact

Problem. Approximately 13-25% of Dutch youth have mental health complaints [3]. With rising depression and anxiety rates the demands on Child and Adolescent Mental Health (CAMH) services increase, and the societal and medical costs have exploded to a staggering 49 billion euro annually in Europe [2]. Even though access to necessary child and adolescent mental health services is a human right, Dutch adolescents wait on average 3-4 years before depression is adequately treated [6].

Cause. The youth mental health-care system is overburdened: Professionals cannot handle the demand, qualified personnel is scarce, waiting lists are long, and the system is no longer financially sustainable.

Underlying cause. To minimize costs and reduce inequality, ideally interventions are offered in a stepped-wise approach, with increasing intensity levels, starting with low-threshold, low-intensity interventions. This requires a major **transition from reactive to proactive care** (or prevention), **from group-based to personalized** care to increase (cost-)effectiveness, and **from hospital to home monitoring and self-management.** We currently lack scientific insights and (cost-)effective solutions to make this major transition towards proactively preventing mental health problems of the young.

From output to impact. This scientific output will be disseminated within and across disciplines (publications, conference, colloquia, media see WP5) but the program will also contribute to achieve a paradigm shift in accepting and using preventive tools for well-being of youngsters and reducing the currently exploding societal costs. The program is designed to generate such societal impact from day 1 onwards (see Figure 2). Societal impact is expected within 10 years of the onset of the program (Year 1-3 scientific insights, Year 3-5 output, > Year 5 outcome).

Valorization strategy

This 5-year flagship program synthesizes innovations at different Technological Readiness Levels and will tilt each of them critical steps further together with external funding and PPS. Examples within the scope of this program include:

- **From TLR 1** (discovering basic principles) studying the causes of mental health problems in WP1, and whether fluctuations in emotions can help to detect emergent depression in scientific settings **to TRL3** new prototypes designed based on these scientific insights. Co-funding with research grants (e.g., VIDI, ERC, ZonMw)
- **From TLR1** (basic principles) of building algorithms for decision making in WP2 **to** TRL7 (prototype is being used in the region). Co-funding by the Municipality of Rotterdam.
- **From TRL4** (proof of concept established) with regard to how eHealth can effectively reduce mental health problems of the young (e.g., Grow It! WP3) **to TRL6**: improved motivational designs are implemented in five academic hospitals as part of specialized care. Co-funding by partners of eHealthJr.

In Year 4, we will start a second cycle of implementation from demonstration of the systems to deployment into the market (i.e., health(care) system; schools, municipalities) as a stable business proposition.

- **From** TRL 7 (prototype system verified) piloted platforms and tools (e.g., Grow It! eHealth, decision tools) will be tilted **to TLR 9**: Dutch youth will have access to proactive innovative solutions to promote well-being. Co-funding by NWA for a PROTECt ME application in Year 4 together with business and societal partners, and Erasmus Research Services.

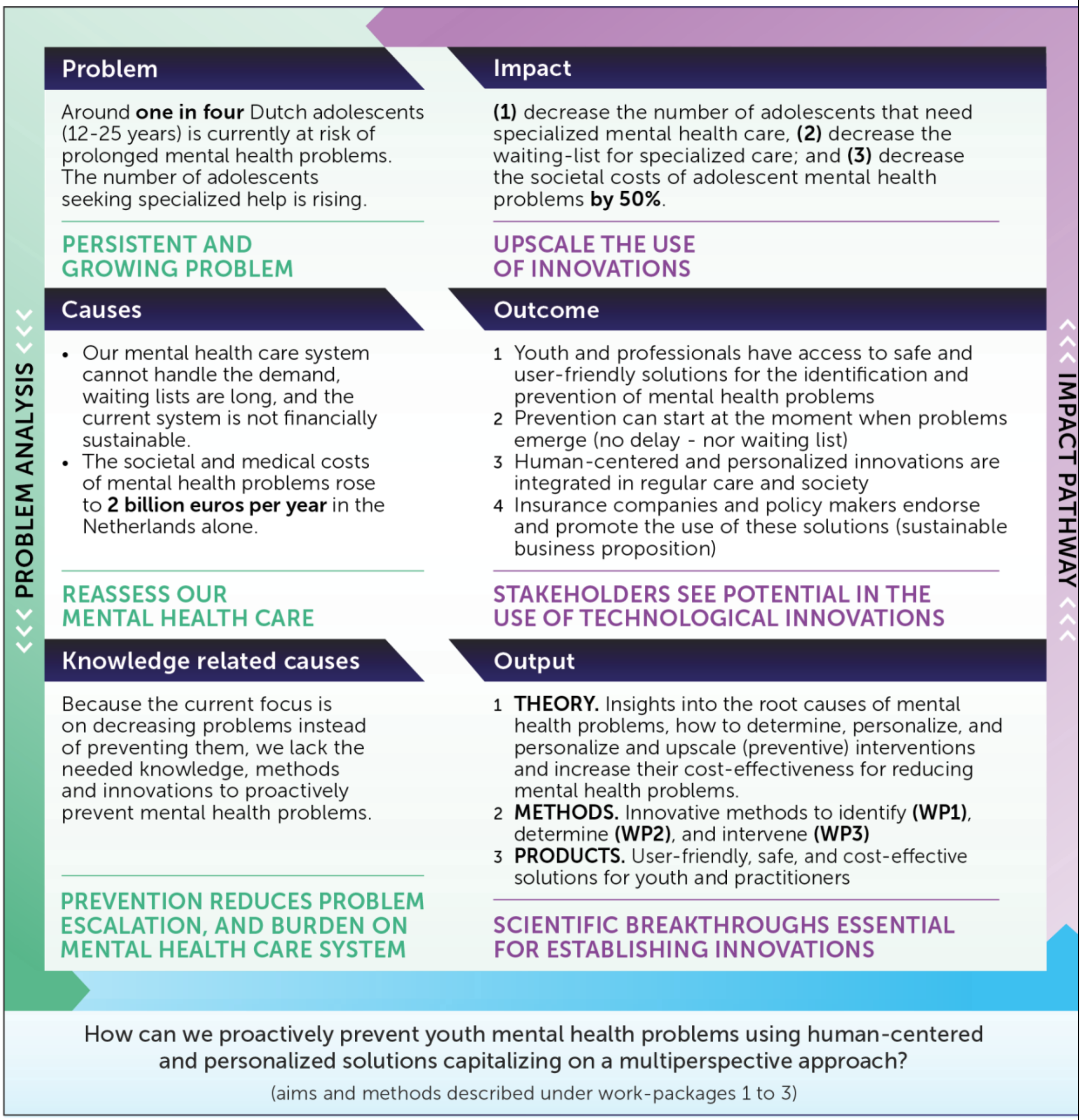

Figure 2. From societal problem to sustainable solutions: Theory of change

3.Flagship proposal (1853 words)

3.a Ambitions

The flagship program is innovative and challenging in its research and approach by addressing **three grand scientific challenges** that currently hinder adequate proactive identification, decision making regarding, and intervention for, mental health problems (WP1-3) while involving stakeholders (WP4) and building sustainable impact (WP5) (see Figure 3). Specific scientific innovations are described per WP in Section 3b.

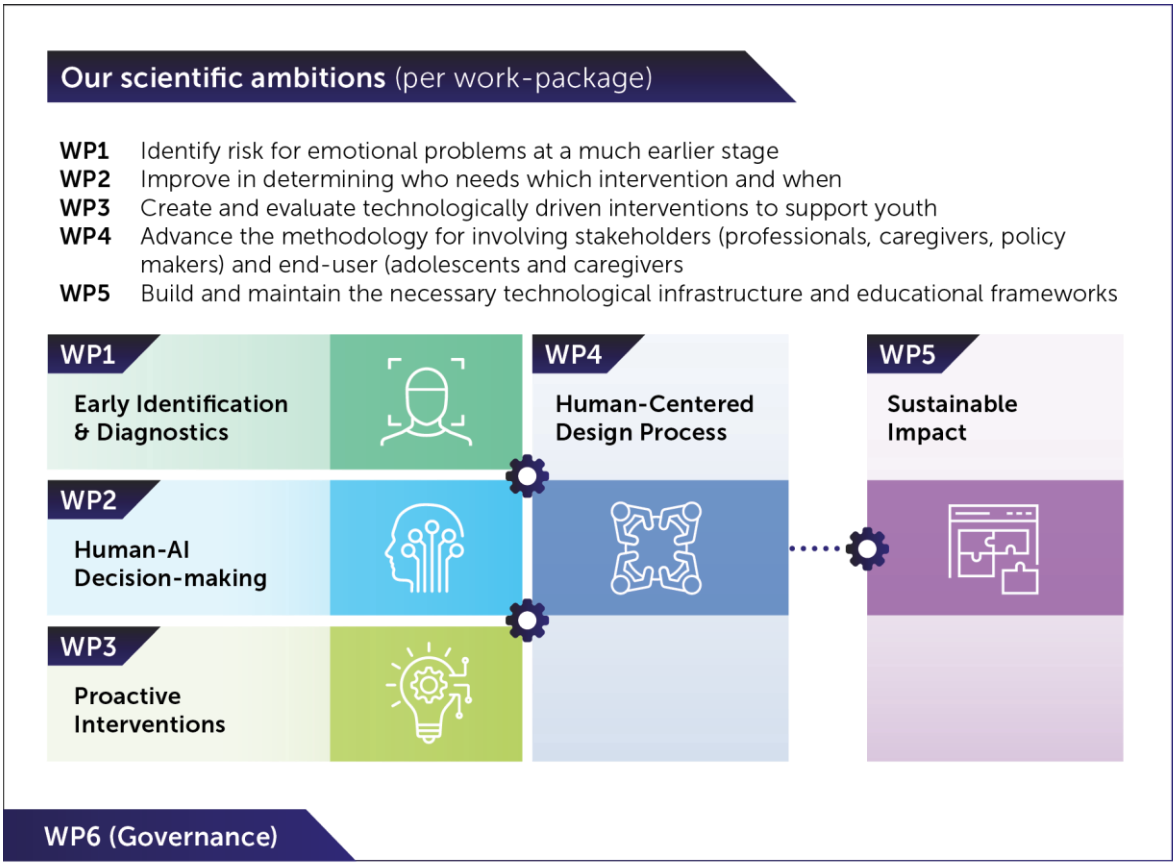

Figure 3. Ambitions and coherence of work packages.

3.b Aims and feasibility

Here we present the concrete aims, plans, tasks and needs per WP. ©refers to co-funded activity. Deliverables: SP=Scientific Paper (1 annually per young scholar). T=Tool (i.e., prototype, new concept, algorithm, intervention, product). M=Milestone, a steppingstone that must be reached in order to continue.

WP1: Early Identification & Diagnostics

Aim. Identify risk for emotional problems at a much earlier stage
Innovation. Combining cutting edge real-time measures and technology, such as Experience Sampling of real-time emotions, with state-of-the-art data science and statistics
Consortium members(7). Vreeker(Erasmus MC; WP-lead), Garcia-Gomez(EUR; WP co-lead), Keijsers(EUR), Gadiraju(TUD), 1 Post-doc, 2 PhD-students
Societal Partners. Ethica data, Generation R, Innovattic
Activities

- 1.1©. Structured review and theoretical synthesis on risk factors (M1.1; SP1-2);
- 1.2. Verifying static risk factor model in administrative data covering full Dutch population to assess risk groups, e.g., poverty, parental job loss (SP3);
- 1.3©. Identifying dynamic risk factors in daily live (SP4) among general population (SP5; GenerationR) and high-risk samples (SP6);
- 1.4. Out-of-sample tests of these risk factors on new data, *n*=300, *t*=120 moments (SP7-8)
- 1.5. Applying machine learning to obtain nomothetic (SP6) and person-specific insights (SP9) into early identification of emotional problems;
- 1.6©. Linking passively logged digital data (e.g., wearables) to predict the onset (SP10) and real-time deterioration of emotional well-being (SP11);

Output. M1.1: Theoretical framework for *PROTECt ME* (M1.1), 11 publications.
Link to other work-packages. Theoretical framework needed for WP2; Requires infrastructure of WP5.
Budget = 677k
<u>Personnel</u>
Post-Doc1
PhD-student1
PhD-student2
= 514k
<u>Materials</u>
1.2: CBS-data(50k)
1.4: 100 wearables(100k), 10 phones(5k), money-rewards(34k), Ethica software(WP5)
= 139k
<u>Third:</u>
External advice (25k)
Existing co-funding©.
1.1/1.3; MARIO(75k)
1.3: Innovatic (25k)
1.3/1.6: eHealthJr (175k)

WP2:Human-AI Decision-making

Aim. Improve decision making about who needs which intervention and when
Innovation. Combining professional expertise with the power of novel Human-AI collaborative workflows.
Consortium members (13). Gadiraju(TUD; WP-lead), Kudina(TUD), Hillegers(Erasmus MC), Legerstee(Erasmus MC), Brinkman(TUD), 2 Post-docs, 6 PhD-students
Societal partners. Ijsfontein
Activities

- 2.1. Identify values and concerns, through interviews and focus groups with youths and professionals (with WP4; SP1–3; M2.1.)
- 2.2. Build referral tool for adolescents to find useful online resources and proactive (eHealth) interventions (T2.2; M2.2.)
  - 2.2a©. Build adaptive, personalized, responsible and ethical algorithms and implement in Grow It! app (SP4–6)
  - 2.2b©. Implement algorithms in interactive chatbots (T2.2b, SP7–9)
  - 2.2c. Explain algorithmic advice to all stakeholders (adolescents, parents, caregivers, etc.) (SP10–12)
- 2.3©. Build decision tool for professionals for referring youth to the right care at the right moment (T2.3a, M2.3a).
  - 2.3a. Empirical validation of decision tool in Generation R population (SP13).
  - 2.3b. Focus groups/interviews on usability, acceptance and conditions among professionals (e.g., family doctor, youth care, municipality) (SP14)
  - 2.3c. Impact analysis (e.g., surveys to key-stakeholders). (SP15)
  - 2.3d©. Implementation of decision tool in several municipalities (VNG) (SP16; M2.4)

Output. 2.1. requirements(M2.1), 2.2. referral tool for adolescents(M2.3), 2.3. decisions tool for professionals(M2.3), 2.4. implemented across country(M2.4), 16 scientific papers.
Link to other work packages. Needs theoretical insights of WP1, refers to interventions(WP3), stakeholder involvement of WP4, data-management of WP5.
Budget=1065k
<u>Personnel</u>
Post-Doc 2-3
PhD-Student 3-8
Student-Assistants
= 835k
<u>Materials</u>
2.2/2.3. Building tool for adolescents and professional (155k)
<u>To third parties:</u>
2.3. Implementation (50k)
<u>Meetings</u>, travel costs interviews (25k)
Co-funding.
2.2 Ijsfontein(30k)
2.3a-c. Municipality Rotterdam(200k)
2.2a PhD6, Perfect Fit NWO(75k)/NWO ELSA lab (75k)
2.2b. PhD4, NWO Robust (75k)

WP3: Proactive Interventions

Aim. Create and evaluate technologically driven interventions to proactively support youth
Innovation. Designing and evaluating adequate, (cost-)effective, personalized and timely solutions, including gamified eHealth.
Consortium members (9). Hillegers(Erasmus MC; WP-lead), Weeland(EUR), Keijsers(EUR), Garcia-Gomez(EUR), Kudina(TUD), 3 PhD-students, 1 Postdoc
Societal partners(5). IJsfontein; Innovattic; MIND Young; PlusMinus; Student well-being program
Activities

- 3.1. Develop plan based on intervention mapping (SP1; M3.1)
- 3.2. Evaluate the game mechanics, user adherence, and optimizing the user experience and personalization in Grow it! (SP2-3)
- 3.3. Design and evaluate of novel concepts for interpersonal intervention with parents/peers (T2, SP4-8)[c]
- 3.4. Map design patterns for real-time personalized preventive interventions to create across-domain dissemination of results (SP9-10)
- 3.5. Implement successful proactive preventive interventions for youth (in decision tool of WP2) (SP11)
- 3.6 Evaluate of cost-effectiveness of eHealth interventions (SP12-13)

Output. Plan for intervention development (M3.1), Interpersonal eHealth (T1), 13 scientific publications
Link to other work packages. Embeds new decision algorithms by WP2, stakeholder involvement of WP4.
Budget = 991k
Personnel:
PhD9
PhD10
PhD11
PD4
= 934k
Materials:
3.6. payment subjects, travels, data-collection(57k)
Co-funding:
3.4-3.5. eHealthJr(250k)

WP4: Human in the center

Aim. Facilitate stakeholder and end-user participation in all phases of the program.
Innovation. Novel methods for involving stakeholders and end-users.
Consortium members (3). Weeland(EUR, WP lead), Gielen (TUD), 1 PhD-student, student-assistants.
Societal partners (10). Sync Lab/Young Experts; Student wellbeing program, MIND Young, PlusMinus (young); Wetenschapsknooppunt DELFT; Betasteunpunt Zuid-Holland; Jantje Beton; NVvP; @ease; VO raad
Activities

- 4.1. Plan for stakeholder involvement (from co-design to implementation)
- 4.2. Young expert panel (SP1)

- 4.3©. Structured systems thinking sessions with caregivers and mental health care professionals (input for other WPs) (SP2-3)
- 4.4. Co-design sessions with professionals and adolescents (input other WPs) (SP4)
- 4.5© Testing sessions for developed innovations from other WPs with professionals and adolescents
- 4.6. Develop a new toolbox and handbook to facilitate adolescent participation as co-researchers and co-designers (e.g. *generative tools* [16], *Collaborative Design Thinking tools* [17] and *cooperative inquiry)* (T1, SP5)[18]

Output: 5 Scientific publications; 1 Handbook; Open access toolbox (T1).
Link to other work packages. Facilitates WP1-3; implementation in education(WP5).
Budget = 403k
Personnel:
PhD12
Student assistant
= 303k
Materials: e.g., camera's, recorders, materials interactive sessions, reimbursement of adolescents(100k)
Co-funding.
4.3. Plus Minus (1k)
4.5. Mind Young (10k)

WP5. Sustainable Impact: Research Infrastructure & Education

Aim. Build and maintain the necessary research infrastructure and educational frameworks for sustainable impact.
Innovation. Interdisciplinary expertise centra, novel educational programs.
Consortium members(5). Keijsers(EUR; research-lead), Brinkman(TUD education-lead), Specht(TUD/EUR), educational developer, data-manager.
Societal partners(6). Sync Lab, Healthy Start program, RE-CONNECT, Municipality of Rotterdam, EHealthJr, Student wellbeing program.

Activities

- 5.1. Joint supervision BSc/MSc theses & Guest lectures (incl. professionals)
- 5.2©. Joint learning materials (e.g., MOOC for various target groups: student, alumni), minor program BSc, building a virtual adolescent for professional training .
- 5.3. Novel master modules *"Technology for mental Health"* (e.g., MSc Joint Interdisciplinary project)
- 5.4. Student participation in research (see below)
- 5.5©. Collaborative research infrastructure for ESM studies - see also Section 2c

Output. Novel educational modules (5.1-5.3) and collaborative research infrastructure (5.4), ESM infrastructure (T1).
Link to other work packages. WP5 delivers infrastructure for all other WPs
Budget = 506k
Personnel:
Educational developer
Data-manager
PhD13
=230k

Material
MOOC(75k)
ESM infrastructure(200k)
Co-Funding.
5.5.Ethica data(100k)
5.2. Saudi Arabian Cultural Bureau(50k)

Task 5.3 and 5.4. Strategy to involve students. Students will participate for example within existing design-oriented courses of *Design for Interaction*, projects within Delft Design Labs (TU Delft) and within courses of *Orthopedagogiek* (EUR), *Opvoedvraagstukken in een diverse samenleving* (EUR) and *Health Sciences* and *Psychiatry* (Erasmus MC). Within our novel Master "design for mental health" students from all institutions will collaborate in our research with a convergence mindset. The development of these new educational modules is led by the director of the Leiden-Delft-Erasmus Center for Education (Specht).

WP6 is described under section 4d (Governance). **Budget:** programmanager(262k), consortium meetings/consumables(77k). **Total:** 340k

FEASIBILITY

All resources are realistically estimated - see budget table. Risk and mitigation plans are described in section 6.7.

Plan for attracting funding

Future **research funding** will come from highly prestigious personal grants to which this relatively young consortium can still apply (NLD: 3 VENIs, 2 VIDIs, 2 VICIs; Europe: 2 ERCst, 1 ERCcons, 2 ERCadv) and from ZonMw grants. We will collectively apply for one NWA-ORC (approx. 5M), and one Gravitation grant (approx. 20M). A success-rate of 20% in acquiring funding for this plan seems highly realistic for this consortium (see Section 4) and this makes the flagship program economically feasible. One of the key coordinating tasks of our program manager is to facilitate growth as a community and we will support and mentor (young) scholars in grant writing.

To develop towards a **sustainable business proposition**, which aligns with all legal, medical, ethical, data-privacy and security regulations we have built strong coalitions, several of which have promised their support and/or contribution (see “Flagship societal partners”). This will ensure strong PSS, as well as additional external funding.

3.c Work plan

WP teams (please see 3b for details). In each work package (WP), at least two institutes will collaborate in small teams of **3-13** consortium members. The **17** young scholars who will join our consortium will be supervised by PhD supervisors from two institutes and they will collaborate with members of our community with relevant expertise (see Table 1). Figure 4 (Gantt chart) demonstrates the coherence between WPs over time.

Data-sharing and storage infrastructure. WP5 organizes the research infrastructure. We will work together with university data stewards and use SURF cloud facilities for secure data sharing. The data-manager will keep the general data management plan, software sustainability plan, and research ethics plan up-to-date. Furthermore, we will use the CII Best Practices Badge Program checklist for software development. We will use existing tools for Open Science (e.g., Github, Open Science Framework) (please see WP5, Section 2c).

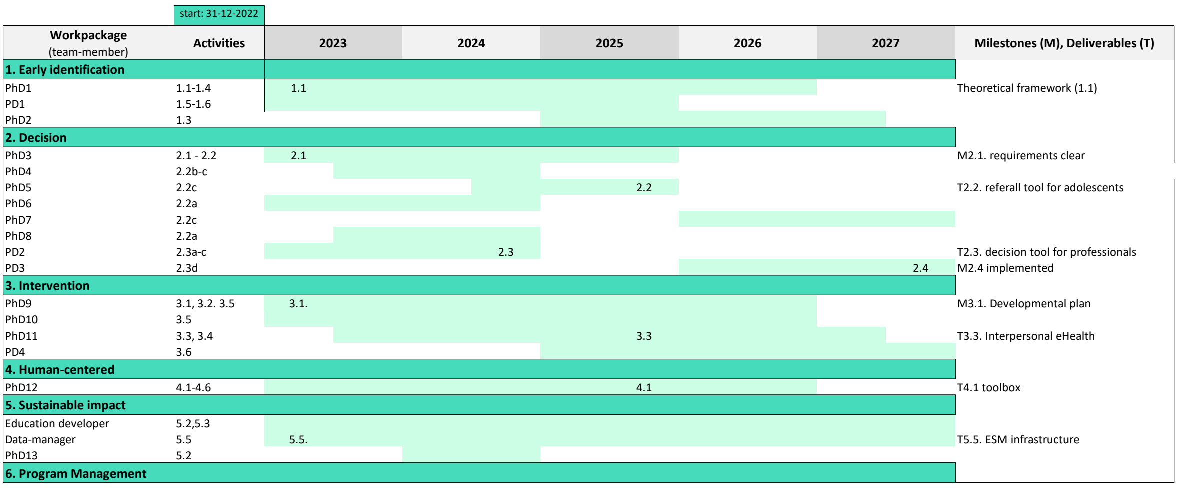

Figure 4. Gantt chart

3.d Alignment of the program with relevant national and international research agenda'

Scientifically, not a single consortium has been able to address mental health problems in the life phase where it most likely emerges: Adolescence.

Within our fields. Capitalizing on progress in each of our disciplines, this interdisciplinary consortium will break new ground by synthesizing innovations in the following domains:

- **create** cutting-edge interventions, which are personalized and timely;
- **evaluate** all solutions on their (cost-)effectiveness;
- **connect** AI to the perspectives of human experts and end-users;
- **deliver** novel methods to monitor daily wellbeing, and algorithms to detect risk in real-time;
- **apply** and improve cutting-edge and inclusive participatory co-design methods; and
- **train** young interdisciplinary academics.

Across domains. Our output is also highly instrumental in tackling other societal challenges (e.g., healthy lifestyle) and/or other populations (e.g., children, adults). Moreover, our new methodologies to proactively monitor well-being in daily lives (e.g., using conversational agents and wearables) can be used in other domains (e.g., organizational psychology, marketing, and product design).

Alignment with research agendas. The program aligns well with ongoing research agendas and programs on interventions, eHealth, and promoting mental health (e.g., eHealthJr; Student well-being program), other convergence initiatives (e.g., Healthy Start). We will, amongst others, valorize on diverse national funding possibilities (e.g., NWA, AI research agenda NWO, National Growth Fund), and the EU's Horizon Europe funding scheme. We have strong European networks in place to achieve our scientific ambition to become a European center of excellence. Section 4d describes our funding strategy.

4. Consortium (1958 words)

4.a Track record of the consortium members

Composition and Diversity of Consortium members. The consortium members are 3 PIs and 8 co-applicants from 6 faculties in 3 institutes. It unites senior (4 full professors) and mid-career scholars (7 assistant/associate professors/postdocs). Our average age is 42 (min=32, max=52). The consortium has a diverse composition: Our female:male ratio is 6:5, we represent 7 disciplines, and members have 5 different countries of origin. After funding, 17 junior scholars (PhD-candidates and postdocs) will join us.

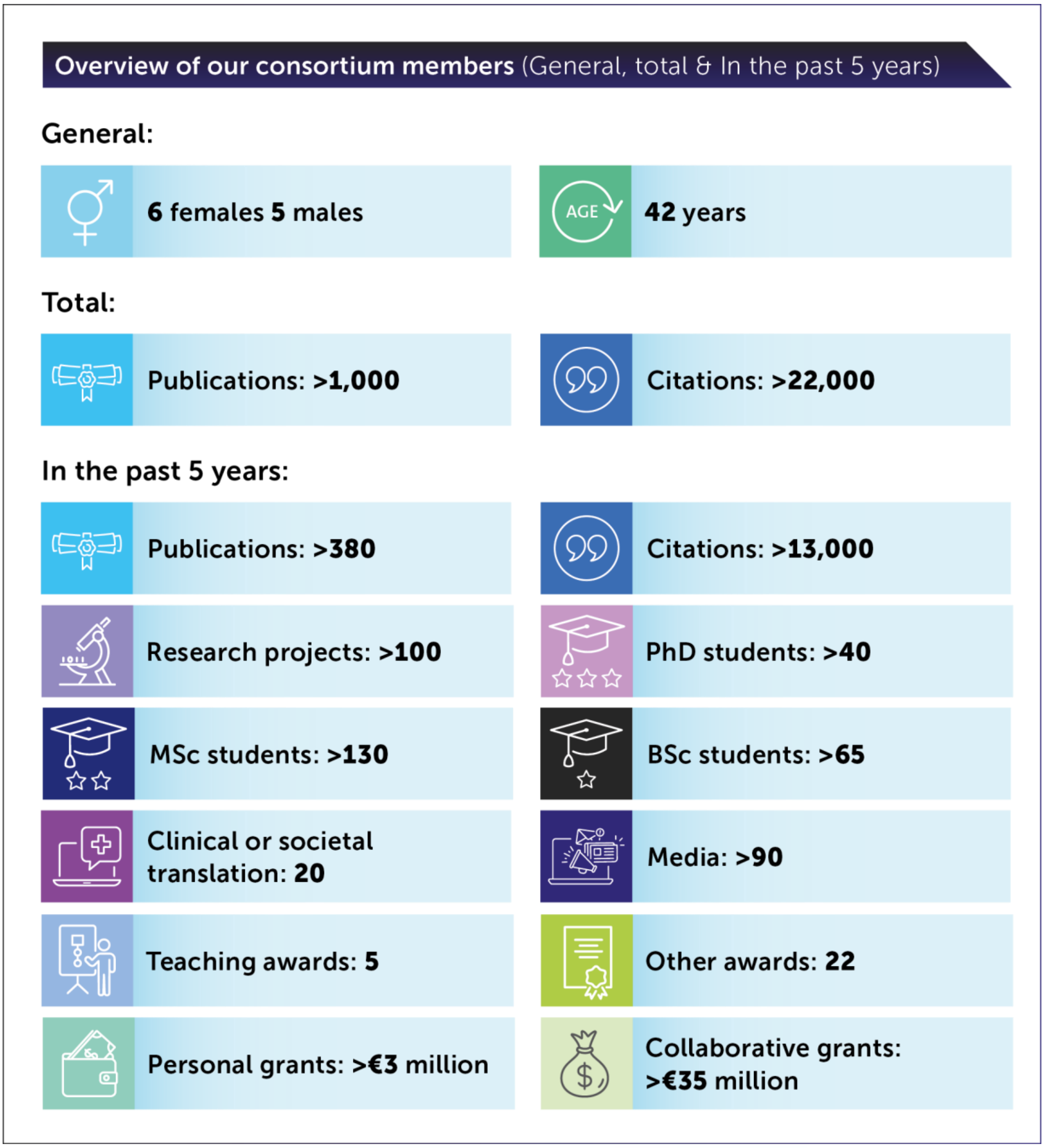

Figure 5. Track record of the 11 applicants.

Track record of supervision. As award-winning, highly productive consortium (Figure 5), we have successfully supervised more than 40 PhD students, many of whom are now successful themselves in academic or societal careers. The main applicant was recently nominated best supervisor by the graduate school, and her PhD-candidate won an award for Open Science. We have jointly mentored around 200 BSc/MSc theses.

Leadership qualities. The three PIs (Keijsers, Hillegers, Brinkman) who have extensive leadership experience with large consortia (e.g., NWA-ORC) and teams/departments (Keijsers chairs a 30+ fte department, Hillegers 220fte and is board member of the Sophia Children's Hospital, Brinkman is Director of Studies MSc Computer Science with 700+ students and 60+ teachers). We have specialized in implementing agile modes of interdisciplinary collaboration, creating inclusive work cultures, and creating societal impact. For instance, the main applicant has recently been awarded the Society for Research on Adolescence interdisciplinary collaborative award. She has successfully implemented agile project, program, and team management in academic settings (up to 50 colleagues) in the last 5 years. As a result, our joint eHealth app Grow It! (Hillegers and Keijsers [19]; Figure 6) was recently played and evaluated by more than 2000 adolescents and has recently been nominated for a computable award.

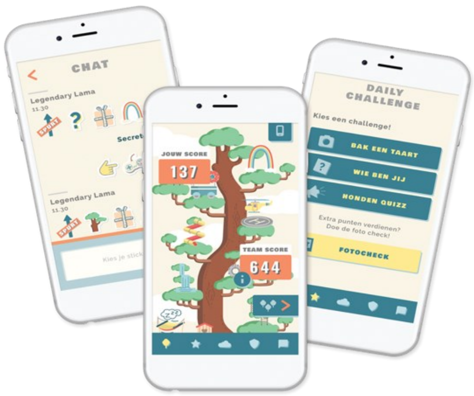

Figure 6. Grow it! app [19]

Potential to obtain funding. This consortium has been highly successful in obtaining external funding. We have obtained 38 million funding in the past 5 years, including NWA-ORC, ZonMw, NWO-VENI and VIDI. More than 51M is awaiting decision (final round), including an ERC consolidator grant, two 20M gravitation grants, one 5M NWA-ORC, one 500k ZonMw grant. Our future funding strategy is aligned with these earlier successes and to obtain 2 to 10M of funding in the next five years is a highly realistic aim.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

Despite our individual successes in terms of output, awards, supervision, and grants, none of us individually have been able to tackle the great challenges described here. This flagship program allows us to converge medical knowledge about mental health(care) (Erasmus MC), social and behavioral sciences (e.g., adolescent development and (micro-econometric analyses of) interventions (EUR), with technological innovations in motivational design and Artificial intelligence (TUD). In Table 1 below we illustrate the complementarity of these disciplines per objective (WP), and the ways in which our partners in the community will support us.

Table 1. Complementarity of scientific disciplines and (societal) partners per objective

WP1. Identify and diagnose risk for emotional problems at a much earlier stage, using cutting edge real-time measures, technology, and data modeling	
Needed Expertise	**Consortium**
Mental health (problems) in youth	**Consortium-members:** Vreeker(ErasmusMC); Keijsers(EUR) **Consortium partners:** van Rossum; Mesman; Lambregtse-van den Berg; Van Haren; Ten Hoopen; Jansen(ErasmusMC); Engels; Laure; Remmerswaal; Crone; Buijzen(EUR) **Societal partners:** PlusMinus; MIND Young
Real-time measures (Experience sampling) / passively logged data; statistical analysis	**Consortium members:** Keijsers(EUR) **Consortium partners:** Specht(EUR/TUDelft) **Societal partners:** EthicaData
Administrative data covering full Dutch population; Health Disparities	**Consortium members:** Garcia-Gomez(EUR) **Consortium partners:** Bago d'Uva; van Ourti (EUR), Erasmus Centre for Data Analytics
Data modeling & machine learning	**Consortium member:** Garcia-Gomez(EUR); Gadiraju(TUD) **Consortium partners:** Jansen(ErasmusMC) **Societal partners:** GenerationR
WP2. Improving decision making of who needs which intervention and when, by combining professional expertise with artificial intelligence	
Needed Expertise	**Consortium**
Human-Centered AI	**Consortium members:** Gadiraju; Brinkman(TUD) **Consortium partner:** Broz, Visch(TUD)
Ethics of Technology	**Consortium members:** Kudina(TUD), Neerincx(TUD) **Consortium partners:** Heine(EUR)
Psychological and Psychiatric Practice; Professionals & interventions	**Consortium members:** Hillegers, Legerstee(ErasmusMC) **Consortium partners**: Dierckx(ErasmusMC), Hakkaart-van Roijen; Ten Hoopen(EUR) **Societal partners:** NVvP; PlusMinus; MIND Young
Building application to monitor emotions	**Consortium members:** Keijsers(EUR); Hillegers(ErasmusMC)

	Consortium partners: Specht(TUD) **Societal partners:** Ijsfontein; Innovatic; Ethica Data
User-centered design	**Consortium members:** Gielen; Brinkman(TUD) **Consortium partners:** Stappers; Visch(TUD); van Buuren(EUR) **Societal partners:** Ijsfontein; Innovattic; eHealth junior
WP3. Creating and testing new technology-driven interventions	
Needed Expertise	**Consortium**
Preventive interventions; eHealth	**Consortium members:** Hillegers(ErasmusMC); Keijsers(EUR); Weeland(EUR) **Consortium partners:** **Societal partners:** IJsfontein; Innovattic
From theory to intervention; Co-design with youth	**Consortium members:** Weeland(EUR); Gielen(TUD; as part of WP4) **Consortium partners:** SyncLab/Young Xperts **Societal partners:** MIND Young, PlusMinus; NVvP; @ease; VO raad
(Cost-)effectiveness analysis	**Consortium members:** Garcia-Gomez(EUR) **Consortium partners:** Hakkaart-van Roijen; Gruber; Bago d'Uva; Van Ourti(EUR); Neerincx(TUD) **Societal partner:** Student well-being program
Ethical and legal aspects	**Consortium members:** Kudina(TUD) **Consortium partner:** Heine(EUR)
WP4. Stakeholder and end-user involvement	
Needed Expertise	**Consortium**
Stakeholder Involvement and co-design with adolescents.	**Consortium members:** Weeland(EUR); Gielen(TUD) **Consortium partners:** Stappers(TUD)**,** van Buuren(EUR) **Societal partners:** Sync Lab/Young Experts; Student wellbeing program, MIND Young, PlusMinus (young); Wetenschapsknooppunt DELFT; Betasteunpunt Zuid-Holland; Jantje Beton; NVVP; @ease; VO raad

WP5. Sustainable impact: Research Infrastructure & Education	
Needed Expertise	**Consortium**
Impact assessment	**Consortium members:** Keijsers(EUR);Brinkman(TUD) **Consortium partners:** Van Buuren(EUR); van der Heiden(EUR) **Societal partners:** Living labs (Sync Lab); Healthy Start program; RE-CONNECT; Municipality of Rotterdam; Student wellbeing program EHealthJr
Innovation in education	**Consortium members:** Brinkman(TUD), Keijsers(EUR), Specht(TUD) **Societal partners:** University educational programs; VO raad

4.c Involvement of partners

Vision. We strive for a transformative approach to stakeholder involvement in which a) we critically consider power relations, b) give voice to marginalized or underrepresented groups, c) use an iterative research process in which we recognize different forms of knowledge and d) strive for mutual understanding and learning. In each stage different forms of involvement are used, for example: (Young) expert panels will be used to gather experience and practical knowledge, co-design techniques will be used to involve stakeholders in the design of innovations, and stakeholders and end-users will be invited to test innovations and their implementation.

Early stages of involvement. Stakeholders and end-users are involved as partners in each stage of the project, from drafting this proposal (Figure 7, as examples) to successful execution of the program (Table 4b), for instance, through providing input (WP1 and 2), designing and testing new interventions (WP3) and implementing our tools/interventions (WP5). Throughout we will iteratively adapt our strategy and plan to align with their values, needs, and requirements.

Figure 7. Above: Co-design sessions (within TUD Master course). Below: Focus group on mental wellbeing with youth resulting in illustrations by Bart Aalbers illustraties.

4.d Governance, organization and collaboration strategy of the consortium

Vision. As seasoned academic leaders, we are well aware of the complexity of program management. To ensure effective and timely decision-making and delivery of outputs, we adopt the management vision of agile working as this is most suited for delivering the right insights, outcomes, and products in complex, iterative environments like ours. Within this program, the agile teams (WPs) will operate as an integrated, evolving systems, with strong iterative alliances across teams (the other WPs) and with end-user and stakeholders' input as driving force (facilitated by WP4). These are the activities and measures that we will undertake to achieve this iterative collaborative workflow and to maximize value and impact for science and society.

WP6. Governance

Aim. A strong governance and agile organization of this consortium to facilitate synergy/value creation.
Consortium members. Keijsers (EUR), Brinkman (TUD), Hillegers (Erasmus MC), program-manager
Activities

- 6.1. Collaborative agreements / quality indicators established / online dashboard
- 6.2. Communication strategy & Impact plan based on Theory of change
- 6.3. Datamanagement plan
- 6.4. Talent management & diversity plan (e.g., scouting, hiring plan, mentorship)
- 6.5. Young Board
- 6.6. Advisory board (scientific and societal)
- 6.7. Risk monitoring and mitigation
- 6.8. Organizing (cross-)team meetings, colloquia, stakeholder involvement
- 6.9. Outreach (e.g., for schools, parents, children) and scholars' training therein
- 6.10. Grant writing strategy: mentorship, skill building, consortium building for new projects

Team members. The team is led by a small daily **board** (3 consortium PIs and 4 work package team leaders) who will meet monthly to discuss progress against mutually agreed quality indicators (6.1), to optimize the infrastructure, to attract new talent, and to seek additional possibilities for convergence (e.g., writing grant applications). A **program manager** (agile program management black belt) will shape our organizational structures and workflows, and support operational activities, such as meetings, building an online dashboard for quality monitoring and real-time iterative feedback, and drafting agreements and strategies.

6.1-6.3. Agreements/plans. The consortium will establish a collaborative agreement (6.1) based on shared values in alignment with the convergence board. During the program, each partners' efforts are coordinated in compliance with this agreement, and our communication/ impact (6.2) and data-management plans (6.3). Quality indicators include:

- Scientific output quantity and quality (e.g., Open Science, Education, Impact indicators, joint PhD-projects);
- Education and societal impact (e.g., novel courses and evaluation, prototypes and user-evaluation, IPs);
- Human resources (e.g., attracting new talents and diversity of consortium, (young) researchers' well-being and perceived quality of supervision);
- Potential for growth (e.g., submitted and acquired grants).

6.4/ 6.5. Talent management program. A **young board** (Phd-students, postdocs) will organize additional activities to converge, and provide the daily board with input on how to optimize interdisciplinary working conditions. Each PhD-student and postdoc will be **supervised** by two seniors (across disciplines). Mid-career scholars will be offered a cross-disciplinary personal **mentorship** by full professors, from the consortium and the Convergence community. Each junior and mid-career researcher will annually discuss their potential career challenges and their **Personal Development Plan** to develop scientific, technical, managerial and leadership skills. With their assigned **peer group**, researchers will share and discuss their career choices, challenges, and opportunities more informally.

6.6. The advisory board will be composed of international experts, and national stakeholders to critically advise the program on the direction, academic innovation, and potential for societal impact. Their advice will also be critically reflected upon by adolescents and parents as part of WP5.

6.7 Monitoring risk and mitigation. In our board meetings, we will evaluate the quality of the established coherent cooperative workflow, and the quality of the interdisciplinary output in our scientific fields. Table 2 gives an overview of the risks currently identified.

Table 2. Risk and mitigation

Risk	Level	Mitigation plan
Involving adolescents, esp. those who are underrepresented, and professionals	*high*	WP4: Adolescents' involvement. Existing collaborations (high schools, patient organizations, youth social workers). Sufficiënt budget.
Data management/privacy/ethics/legal	*high*	Board member has this as a dedicated task. Experts in community and advisory board. Solid data-management plan. Alignment with ethical standards within institutes (e.g., Helskinki). (Medical) ethical evaluation of each data-collection beforehand. Advice from privacy officers and data stewards.
Diversity of consortium	*middle*	Talent management plan; Advice from diversity officer.
Societal impact	*middle*	Stakeholder involvement, strong coalitions with societal and business partners. Involvement of Arwin van Buuren (impact-lead EUR); Monitor half-yearly and improve plan accordingly; Agile governance vision
Practical hurdles	*middle*	Ongoing conversations with the board of three institutes regarding needed facilities. Conflict management strategy (WP6). Black belt program manager. Input from young board.
Educational embedding	*middle*	Involving program directors and deans early on, starting with pilots(WP5); a dedicated team-member.
Insufficiënt co-funding	*low*	Grant writing plan: Training, writing workshops, writing weeks, mentorship.

6.8 Collaboration. To optimally use and share expertise we will organize cross-team activities: Half-yearly symposia for all team-members, colloquia with external keynotes, joint supervision on all PhD- and postdoc projects from at least two institutes, and peer-to-peer skill-building sessions.

6.10. Grant writing strategy. Young and mid-career scholars will successfully apply for grants, through mentorship, skill building courses, and consortium building. We will further organize half-yearly brainstorming and writing sessions in order to obtain future co-funding, make this consortium grow and ensure a lasting positive impact on the mental health of youth.

I Budget table (including the external funding table)

Budget formulier: Convergentie Flagship Call

Flagship:	Technology & Health	
Flagship acronym:	PROTECT Me	
Naam Penvoerder:	Loes Keijsers	
Startdatum en looptijd van het project:	31-12-2022	31-12-2027
Totaal project budget:	4,000,000	

!!!
In de onderstaande tabellen 1 t/6 betreft 'in cash' de directe financiering vanuit het flagship.
De 'in kind' kosten betreft de inbreng middels externe projecten.

R&D Overview: kostensoorten per type onderzoek (hele project periode)

1.

Loonkosten:						2023		2024		2025		2026		2027		Totaal			Totaal per partner
Partner die de kosten maakt	Functie	fte	Inschaling	Maandbedrag	Totale kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind		
Onderzoeksorganisaties																			
Erasmus Universiteit (ESSB & ESE)	PhD student	€ 1.00	VSNU P0-P3		€ 253,000.00	€ 63,250.00		€ 63,250		€ 63,250		€ 63,250				€ 253,000	€ -	EUR	
Erasmus Universiteit (ESSB & ESE)	PhD student	€ 1.00	VSNU P0-P3		€ 253,000.00		€ 63,250		€ 63,250		€ 63,250		€ 63,250			€ -	€ 253,000		€ 1,030,500
Erasmus Universiteit (ESSB & ESE)	Post doc	€ 1.00	VSNU 11.0-11.1		€ 168,000.00				€ 84,000		€ 84,000					€ -	€ 168,000		
Erasmus Universiteit (ESSB & ESE)	Programm manager	€ 0.80	VSNU 10.0-10.4		€ 262,500.00	€ 52,500.00		€ 52,500		€ 52,500		€ 52,500		€ 52,500		€ 262,500	€ -		
Erasmus Universiteit (ESSB & ESE)	Data steward	€ 0.20	VSNU 11.12		€ 94,000.00	€ 18,800.00		€ 18,800		€ 18,800		€ 18,800		€ 18,800		€ 94,000	€ -	EMC	
Erasmus Medical Center (Child & Youth Psychiatry)	PhD student	€ 1.00	Schaal 10.0		€ 262,500.00	€ 64,375.00		€ 64,375		€ 64,375		€ 69,375				€ 262,500	€ -		€ 1,225,000
Erasmus Medical Center (Child & Youth Psychiatry)	PhD student	€ 1.00	Schaal 10.0		€ 262,500.00	€ 64,375.00		€ 64,375		€ 64,375		€ 69,375				€ 262,500	€ -		
Erasmus Medical Center (Child & Youth Psychiatry)	PhD student		Schaal 10.0		€ 175,000.00	€ 87,500.00		€ 87,500								€ 175,000	€ -		
Erasmus Medical Center (Child & Youth Psychiatry)	Post doc		Schaal 10.6		€ 75,000.00	€ 75,000.00										€ 75,000	€ -	TUD	
Erasmus Medical Center (Child & Youth Psychiatry)	Post doc		Schaal 10.6		€ 100,000.00	€ 50,000.00		€ 50,000								€ 100,000	€ -		€ 821,500
Erasmus Medical Center (Child & Youth Psychiatry)	Post doc		Schaal 10.6		€ 100,000.00					€ 50,000		€ 50,000				€ 100,000	€ -		
Erasmus Medical Center (Child & Youth Psychiatry)	PhD student		Schaal 10.0		€ 250,000.00	€ 62,500.00		€ 62,500		€ 62,500		€ 62,500				€ 250,000	€ -		
TU Delft (EWI & IO)	PhD student	€ 1.00	VSNU P0-P3		€ 250,000.00	€ 62,500.00		€ 62,500		€ 62,500		€ 62,500				€ 250,000	€ -		
TU Delft (EWI & IO)	PhD student		VSNU P0-P3		€ 75,000.00		€ 37,500		€ 37,500							€ -	€ 75,000		
TU Delft (EWI & IO)	PhD student		VSNU P0-P3		€ 80,000.00						€ 40,000		€ 40,000			€ -	€ 80,000		
TU Delft (EWI & IO)	PhD student		VSNU P0-P3		€ 75,000.00		€ 37,500		€ 37,500							€ -	€ 75,000		
TU Delft (EWI & IO)	PhD student		VSNU P0-P3		€ 80,000.00						€ 40,000		€ 40,000			€ -	€ 80,000		
TU Delft (EWI & IO)	PhD student		VSNU P0-P3		€ 75,000.00		€ 37,500		€ 37,500							€ -	€ 75,000		
TU Delft (EWI & IO)	Student assistant				€ 50,000.00	€ 10,000.00		€ 10,000		€ 10,000		€ 10,000		€ 10,000		€ 50,000	€ -		
TU Delft (EWI & IO)	PhD student		VSNU P0-P3		€ 50,000.00		€ 25,000		€ 25,000							€ -	€ 50,000		
TU Delft (EWI & IO)	Education developer	€ 0.30	VSNU 11.12		€ 86,500.00					€ 28,833		€ 28,833		€ 28,833		€ 86,500	€ -		
				Totaal	€ 3,077,000													Totaal loonkosten	€ 3,077,000

2.

Kosten van materialen en hulpmiddelen:				2023		2024		2025		2026		2027		Totaal			Totaal per partner
Partner die de kosten maakt	Materiaal / hulpmiddel		Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind		
Onderzoeksorganisaties																	
Erasmus Universiteit (ESSB & ESE)	Kosten van materialen en hulpmiddelen: CBS data (wp 1)		€ 50,000	€ 10,000		€ 10,000		€ 10,000		€ 10,000		€ 10,000		€ 50,000	€ -	EUR	
Erasmus Medical Center (Child & Youth Psychiatry)	Kosten van materialen en hulpmiddelen: Wearables WP1		€ 89,000					€ 29,667		€ 29,667		€ 29,667		€ 89,000	€ -		€ 427,000
Erasmus Medical Center (Child & Youth Psychiatry)	Kosten van materialen en hulpmiddelen: WP2		€ 30,000		€ 15,000		€ 15,000							€ -	€ 30,000		
Erasmus Medical Center (Child & Youth Psychiatry)	Kosten van materialen en hulpmiddelen: WP2		€ 75,000	€ 37,500		€ 37,500								€ 75,000	€ -		
Erasmus Medical Center (Child & Youth Psychiatry)	Kosten Overig WP2		€ 25,000	€ 12,500		€ 12,500								€ 25,000	€ -		
Erasmus Medical Center (Child & Youth Psychiatry)	Kosten van materialen en hulpmiddelen: WP2		€ 50,000	€ 10,000		€ 10,000		€ 10,000		€ 10,000		€ 10,000		€ 50,000	€ -		
Erasmus Medical Center (Child & Youth Psychiatry)	Kosten van materialen en hulpmiddelen: WP3		€ 57,000	€ 11,400		€ 11,400		€ 11,400		€ 11,400		€ 11,400		€ 57,000	€ -	EMC	
Erasmus Universiteit (ESSB & ESE)	Kosten van materialen en hulpmiddelen: stakeholder sessions/ materials (wp 4)		€ 100,000	€ 25,000		€ 25,000			€ 25,000		€ 25,000			€ 50,000	€ 50,000		€ 326,000
Erasmus Universiteit (ESSB & ESE)	Infrastructure ESM: Ethica data wp5		€ 200,000	€ 20,000	€ 20,000	€ 20,000	€ 20,000	€ 20,000	€ 20,000	€ 20,000	€ 20,000	€ 20,000	€ 20,000	€ 100,000	€ 100,000		
Erasmus Universiteit (ESSB & ESE)	Meetings, consumables etc.		€ 77,000	€ 15,400		€ 15,400		€ 15,400		€ 15,400		€ 15,400		€ 77,000	€ -	TUD	
Ondernemingen																	€ 0
Onderneming 2																	
Onderneming 3																	
Onderneming 4																	
Overige (private) partners																	
Naam 1																	
Naam 2																	
Naam 3																	
Naam 4																	
		Totaal	€ 753,000													Totaal kosten materialen en hulpmiddelen	€ 753,000

Application form Flagship program round 2021-2022

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3.

Kosten van gebruik van machines en apparatuur					2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt	Machine / apparatuur	Prijs per gebruikseenheid	Gebruiks eenheden	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																	
Organisatie 1																	
Organisatie 2																	
Organisatie 3																	
Organisatie 4																	
Ondernemingen																	
Onderneming 1																	
Onderneming 2																	
Onderneming 3																	
Onderneming 4																	
Overige (private) partners																	
Naam 1																	
Naam 2																	
Naam 3																	
Naam 4																	
			Totaal	**€ 0**												**Totaal kosten machines en apparatuur**	**€ 0**

4.

Aan derden verschuldigde kosten				2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt	Naam derde, omschrijving kosten		Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties																
Erasmus Medical Center (Child & Youth Psychiatry)	VNG: WP2		€ 50,000					€ 16,666.67		€ 16,667		€ 16,666.67		€ 50,000		**EUR**
Erasmus Medical Center (Child & Youth Psychiatry)	Kosten advisory WP1		€ 25,000	€ 12,500.00		€ 12,500								€ 25,000		
TU Delft (EWI & IO)	MOOC (wp 5)		€ 75,000							€ 75,000				€ 75,000		**EMC**
Organisatie 4																€ 75,000
Ondernemingen																**TUD**
Onderneming 1																€ 75,000
Onderneming 2																
Onderneming 3																
Onderneming 4																
Overige (private) partners																
Naam 1																
Naam 2																
Naam 3																
Naam 4																
		Totaal	**€ 150,000**												**Totaal kosten derden**	**€ 150,000**

6.

Totale kosten	**€ 4,000,000**	**€ 4,000,000**

FINANCIERING - Bijdragen per partner per jaar

	2023		2024		2025		2026		2027		Totaal		Totaal per partner
in Euro	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	
Onderzoeksorganisaties													
Co-financiering externe bronnen ingebracht door EMC													
E-health jr. (Hilligers/Keijsers)	€ 250,000		€ 175,000								€ 425,000	€ 0	€ 425,000
MARIO study (Hilligers)	€ 75,000										€ 75,000	€ 0	€ 75,000
Co-financiering externe bronnen ingebracht door EUR													
ERC - future application (Keijsers)	€ 63,250		€ 63,250		€ 63,250		€ 63,250				€ 253,000	€ 0	€ 253,000
NWO VICI/ERC - future application (Garcia)			€ 84,000		€ 84,000						€ 168,000	€ 0	€ 168,000
ZonMw Preventie/VIDI - future application (Weeland)					€ 25,000		€ 25,000				€ 50,000	€ 0	€ 50,000
co-financiering externe bronnen ingebracht door TUD													
NWO Robust (Gadiraju)	€ 37,500		€ 37,500								€ 75,000	€ 0	€ 75,000
NWO - future appl					€ 40,000		€ 40,000				€ 80,000	€ 0	€ 80,000
NWO Perfect fit (Brinkman)	€ 37,500		€ 37,500								€ 75,000	€ 0	€ 75,000
NWO - future appl					€ 40,000		€ 40,000				€ 80,000	€ 0	€ 80,000
NWO Elsa lab (Neerinx)	€ 37,500		€ 37,500								€ 75,000	€ 0	€ 75,000
Saudian Arabian (Brinkman)	€ 25,000		€ 25,000								€ 50,000	€ 0	€ 50,000
Totale bijdrage onderzoeksorganisaties	*€ 525,750*	*€ 0*	*€ 459,750*	*€ 0*	*€ 252,250*	*€ 0*	*€ 168,250*	*€ 0*	*€ 0*	*€ 0*	*€ 1,406,000*	*€ 0*	*€ 1,406,000*
Ondernemingen													
Ethica Data Software	€ 20,000		€ 20,000		€ 20,000		€ 20,000		€ 20,000		€ 100,000	€ 0	€ 100,000
											€ 0	€ 0	€ 0
To be determined (wp 1)				€ 89,000							€ 0	€ 89,000	€ 89,000
Totale bijdrage ondernemingen	*€ 20,000*	*€ 0*	*€ 20,000*	*€ 89,000*	*€ 20,000*	*€ 0*	*€ 20,000*	*€ 0*	*€ 20,000*	*€ 0*	*€ 100,000*	*€ 89,000*	*€ 189,000*
Overige (private) partners													
Gemeente Rotterdam	€ 100,000		€ 100,000								€ 200,000	€ 0	€ 200,000
VNG					€ 50,000		€ 50,000		€ 50,000		€ 150,000	€ 0	€ 150,000
Innovatic		€ 25,000									€ 0	€ 25,000	€ 25,000
Ijsfontein		€ 30,000									€ 0	€ 30,000	€ 30,000
Totale overige bijdrage	*€ 100,000*	*€ 55,000*	*€ 100,000*	*€ 0*	*€ 50,000*	*€ 0*	*€ 50,000*	*€ 0*	*€ 50,000*	*€ 0*	*€ 350,000*	*€ 55,000*	*€ 405,000*
PPS-toeslag / subsidies / sponsors													
											€ 0	€ 0	€ 0
											€ 0	€ 0	€ 0
											€ 0	€ 0	€ 0
Totaal subsidies	*€ 0*	*€ 0*	*€ 0*	*€ 0*	*€ 0*	*€ 0*	*€ 0*	*€ 0*	*€ 0*	*€ 0*	*€ 0*	*€ 0*	*€ 0*
	in cash	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**	**in cash**	**in kind**			**Totaal budget externe financiering**
Totaal budget per jaar	€ 645,750	€ 55,000	€ 579,750	€ 89,000	€ 322,250	€ 0	€ 238,250	€ 0	€ 70,000	€ 0	€ 1,856,000	€ 144,000	€ 2,000,000
													Totaal verzoek financiering Convergence flagship
		-											2,000,000-
Totaal budget	€ 700,750		€ 668,750		€ 322,250		€ 238,250		€ 70,000		€ 2,000,000		

II Letters of intent and support

NEDERLANDSE VERENIGING VOOR PSYCHIATRIE

Erasmus School of Social and Behavioral Sciences
Dean prof. V. Bekkers
Burgemeester Oudlaan 50
3062 PA ROTTERDAM

Utrecht, 10 maart 2022

Kenmerk: CJ/lw/3809/22
Betreft: Statement of support for the PROTECt ME PROJECT

Dear prof. Hillegers, dear prof. Keijsers,

With this letter we declare on behalf of the Netherlands Psychiatric Association (NVvP) our interest in and support for your research proposal "PROactive TEChnology-supported prevention and MEntal health in adolescence" (PROTECt ME) as applied for by the main applicants Manon Hillegers, Professor of Child and Adolescent Psychiatry, Erasmus MC, Rotterdam, The Netherlands and Loes Keijsers, Professor in Clinical Child and Family Studies at the Department of Psychology, Education and Child Studies. Erasmus School of Social and Behavioural Sciences at Erasmus University, Rotterdam. We are very interested in supporting this consortium in its efforts to optimize the mental health of the young participating in this research project.

The Netherlands Psychiatric Association is the professional scientific association of psychiatrists in the Netherlands. More than 3700 psychiatrists (in training) are members of the association. The NVvP is committed to good psychiatric care in the Netherlands and invests in high-quality education, training and the acquisition of scientific knowledge. We are also involved in setting and raising standards of psychiatry and being the voice of psychiatry.

The board of NVvP has taken notice of the PROTECt ME proposal. We consider the early identification and intervention of youth at high risk for mental health problems as an very important topic of research. The societal impact of this project that enhances decision making regarding who needs mental health care, when and where, is essential these coming years with regard to the mental health costs and the scarcity of qualified personnel. Therefore we believe that the overall aim of this PROTECt ME project proposal and NVvP to strive for optimal well-being of all youth, align very well. We are willing to support this project with an active participation of child and adolescent psychiatrists.

Yours sincerely,
On behalf of The Netherlands Psychiatric Association

Elnathan Prinsen
Chair of NVvP

bezoekadres Mercatorlaan 1200, Utrecht *postadres* Postbus 20062, 3502 LB Utrecht
telefoon 030 282 33 03 *email* info@nvvp.net *website* www.nvvp.net

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[Use headed paper of party]

Dean prof. V. Bekkers, Erasmus School of Social and Behavioural Sciences,
Erasmus University, Burg. Oudlaan 50, Rotterdam, The Netherlands

...February 2022

LETTER OF INTENT
for the
PROTECt ME PROJECT

Dear prof. Hillegers and prof. Keijsers,

Dr. Y.T.H.P. van Duijnhoven Director Public Health

I, ..., in my capacity of ..... at Gemeente Rotterdam hereby confirm that Municipality of Rotterdam intends to contribute to the PROactive TECHnology-supported prevention and MEntal health in adolescence (PROTECt ME project), as applied for by the main applicant, Loes Keijsers, Professor in Clinical Child and Family Studies at the Department of Psychology, Education and Child Studies. Erasmus School of Social and Behavioural Sciences at Erasmus University, Rotterdam.

The PROTECt ME project summary:

Societal problem. 13-25% of Dutch youth (age 12-25) suffer from depression or anxiety. Such mental health problems have long lasting negative effects, including educational dropout, worse physical health, and increased inequality. Mental health problems cost our society 2 billion euro per year. Adequate and timely identification and treatment are currently impossible, due to long waiting lists, time consuming monitoring procedures, and absence of widely and easily applicable personalized prevention strategies. Hence, systematic changes in our approach to youth mental health are sorely needed.

Scientific ambition: Optimizing well-being of Dutch youth aged 12-25 years by:

(WP 1) improving early identification of mental health problems, using cutting edge real-time measures, technology, and data modeling;

(WP 2) improving and implementing decision making of who needs which intervention and when, combining professional expertise with artificial intelligence;

(WP 3) creating and testing new technologically-driven interventions (e.g., eHealth and VR).

Letter of Intent for a project partner

Our integrated approach involves stakeholders and end users in each step (WP 4); and builds a sustainable infrastructure and collaboration for convergence and impact (WP 5).

Output, outcome, impact. Our ground-breaking scientific insights into the root causes of mental health will immediately provide new ways to improve detection, decision making and intervention, new methods and approaches, and user-friendly and cost-effective solutions for youth and professionals. In the long-term, this results in proactive, personalized, accessible, and cost-effective prevention of mental health problems, based on a sustainable business proposition. The ultimate impact will be a decrease in adolescents entering specialized mental health care, shorter waiting lists and a significant reduction in societal costs of adolescent mental health problems. But most importantly; optimal well-being of all youth.

Fit to the convergence. Medical sciences, social science and humanities and technology converge to promote grand transitions in mental health care, for those who need us most; Our children, our future. This flagship proposal in in the heart of two Convergence programs: Health&Tech and Healthy Start.

The Municipality of Rotterdam and five senior experts; prof. Manon Hillegers, prof. Hanan El Marroun, prof. Pauline Jansen, prof. Loes Keijsers and prof. Eveline Crone from the Erasmus School of Social and Behavioural Sciences at Erasmus University and Erasmus Medical Center-Sophia, collaborate within the frame work of RE-CONNECT – Optimaal Opgroeien in Rotterdam (rotterdamexperts-connect.nl) and they focus on optimizing mental health of Rotterdam youth. This PROTECt ME project is a result of this collaboration and in this project the municipality of Rotterdam will contribute on the overarching aim to optimize mental health of Rotterdam youth by improving early identification of mental health problems and providing early interventions.

The municipality of Rotterdam intents to provide € 100K in-cash towards the project costs within WP 2 during year 3-5 and an in-kind contribution, recruitment support in youth Hubs, staff to organise workshops with youth, locations for knowledge exchange and youth participation , representing a monetary value of € 100K, in support of joint projects. This is in accordance with the further detailed project proposal and budget form.

Yours sincerely,

Name: Dr. Y.T.H.P. van Duijnhoven

plusminus
leven met bipolariteit

Huis voor de Gezondheid
gebouw 'de Argonaut'
T: 033-3032350

Adres
Stationsplein 125
3818 LE Amersfoort

E bureau@plusminus.nl
W www.plusminus.nl

Datum
1-3-2022

Onderwerp
Letter of Intent

LETTER OF INTENT
for the
PROTECt ME PROJECT

Dear prof. Hillegers and prof. Keijsers,

I Henk Mathijssen in my capacity of charmain at patient association Plusminus hereby confirm that Plusminus intends to contribute to the PROactive TECHnology-supported prevention and MEntal health in adolescence (PROTECt ME project), as applied for by the main applicant, Loes Keijsers, Professor in Clinical Child and Family Studies at the Department of Psychology, Education and Child Studies. Erasmus School of Social and Behavioural Sciences at Erasmus University, Rotterdam in collaboration with Prof. dr. Manon Hillegers, Department of Child and Adolescent Psychiatry/Psychology at the Erasmus MC Sophia, Rotterdam.
The PROTECt ME project summary:
Societal problem. 13-25% of Dutch youth (age 12-25) suffer from depression or anxiety. Such mental health problems have long lasting negative effects, including educational dropout, worse physical health, and increased inequality. Mental health problems cost our society 2 billion euro per year. Adequate and timely identification and treatment are currently impossible, due to long waiting lists, time consuming monitoring procedures, and absence of widely and easily applicable personalized prevention strategies. Hence, systematic changes in our approach to youth mental health are sorely needed.
Scientific ambition: Optimizing well-being of Dutch youth aged 12-25 years by:
(WP 1) improving early identification of mental health problems, using cutting edge real-time measures, technology, and data modeling;
(WP 2) improving and implementing decision making of who needs which intervention and when, combining professional expertise with artificial intelligence;
(WP 3) creating and testing new technologically-driven interventions (e.g., eHealth and VR).
Our integrated approach involves stakeholders and end users in each step (WP 4); and builds a sustainable infrastructure and collaboration for convergence and impact (WP 5).
Output, outcome, impact. Our ground-breaking scientific insights into the root causes of mental health will immediately provide new ways to improve detection, decision making and intervention,

plusminus
leven met bipolariteit

Datum
1 maart 2022

Pagina
2/2

new methods and approaches, and user-friendly and cost-effective solutions for youth and professionals. In the long-term, this results in proactive, personalized, accessible, and cost-effective prevention of mental health problems, based on a sustainable business proposition. The ultimate impact will be a decrease in adolescents entering specialized mental health care, shorter waiting lists and a significant reduction in societal costs of adolescent mental health problems. But most importantly; optimal well-being of all youth.

Fit to the convergence. Medical sciences, social science and humanities and technology converge to promote grand transitions in mental health care, for those who need us most; Our children, our future.

Plusminus intends to provide an in-kind contribution in order to support focus groups for youth with (risk of) mental illness, co-creation in research, and dissemination of results, representing a monetary value of € 1040,00 and further detailed in the project proposal and budget form.

	(Cost-) price or rate	*Amount*
Facilitating and participating in focus groups (6 hours)	€65/hour	€390
Co-creation in research projects (6 hours)	€65/hour	€390
Dissemination of results (4 hours)	€65/hour	€260
Total estimated		€1040

Yours sincerely,

Name: Henk Mathijssen

Position: Chairman Plusminus

Date: 1-3-2022

Dean prof. V. Bekkers, Erasmus School of Social and Behavioural Sciences,
Erasmus University, Burg. Oudlaan 50, Rotterdam, The Netherlands

2 March 2022

LETTER OF INTENT

for the

PROTECt ME PROJECT

Dear prof. Hillegers and prof. Keijsers,

I, Hans Luyckx, in my capacity of board member at IJsfontein hereby confirm that IJsfontein b.v. intends to contribute to the PROactive TECHnology-supported prevention and MEntal health in adolescence (PROTECt ME project), as applied for by the main applicant, Loes Keijsers, Professor in Clinical Child and Family Studies at the Department of Psychology, Education and Child Studies. Erasmus School of Social and Behavioural Sciences at Erasmus University, Rotterdam.

The PROTECt ME project summary:

Societal problem. 13-25% of Dutch youth (age 12-25) suffer from depression or anxiety. Such mental health problems have long lasting negative effects, including educational dropout, worse physical health, and increased inequality. Mental health problems cost our society 2 billion euro per year. Adequate and timely identification and treatment are currently impossible, due to long waiting lists, time consuming monitoring procedures, and absence of widely and easily applicable personalized prevention strategies. Hence, systematic changes in our approach to youth mental health are sorely needed.

Scientific ambition: Optimizing well-being of Dutch youth aged 12-25 years by:

(WP 1) improving early identification of mental health problems, using cutting edge real-time measures, technology, and data modeling;

(WP 2) improving and implementing decision making of who needs which intervention and when, combining professional expertise with artificial intelligence;

(WP 3) creating and testing new technologically-driven interventions (e.g., eHealth and VR).

Our integrated approach involves stakeholders and end users in each step (WP 4); and builds a sustainable infrastructure and collaboration for convergence and impact (WP 5).

Output, outcome, impact. Our ground-breaking scientific insights into the root causes of mental health will immediately provide new ways to improve detection, decision making and intervention, new methods and approaches, and user-friendly and cost-effective solutions for youth and professionals. In the long-term, this results in proactive, personalized, accessible, and cost-effective prevention of mental health problems, based on a sustainable business proposition. The ultimate impact will be a decrease in adolescents entering specialized mental health care, shorter waiting lists and a significant reduction in societal costs of adolescent mental health problems. But most importantly; optimal well-being of all youth.

Fit to the convergence. Medical sciences, social science and humanities and technology converge to promote grand transitions in mental health care, for those who need us most; Our children, our future.

IJsfontein b.v. intends to provide an in-kind contribution of building and implementing the machine learning personalised profiles in the Grow it! App, representing a monetary value of € 30.000,- and further detailed in the project proposal and budget form.

Yours sincerely,

Name: Hans Luyckx

Position: Board Member

Date: 2 March 2022

Dear Prof. V. Bekkers, Erasmus School of Social and Behavioral Sciences, Erasmus University, Burg. Oudlaan 50, Rotterdam The Netherlands

10 March 2022

LETTER OF INTENT
for the

Protect me

PROJECT

Dear Prof. Bekkers, Dear Prof. Keijsers,

I, Amin Tavassolian, in my capacity of Senior Product Manager at Ethica Data Services Inc., hereby confirm that Ethica Data intends to contribute to the Protect Me project, as applied for by the main applicant, Loes Keijsers, Full Professor at the Erasmus University Rotterdam, The Netherlands.

Ethica Data intends to provide an in-kind contribution over the five years of the program, composed building and maintaining local infrastructure and hosting for researchers to collect *Experience Sampling Data* with our application representing a monetary value of € 100.000,- and further detailed in the description of Work package 5, Annex 1 of this letter, and in the project proposal and budget form.

Yours sincerely,

Name: Amin Tavassolian

Position: Senior Product Manager

Date: 10 March 2022

MIND Platform

Erasmus University
Erasmus School of Social and Behavioural Sciences
Attn. Dean prof. V. Bekkers
Burg. Oudlaan 50
Rotterdam, The Netherlands

Date	**:**	March 8, 2022
Kenmerk	**:**	22-010
Subject	**:**	**LETTER OF INTENT for the project PROTECt ME**

Dear prof. Hillegers and prof. Keijsers,

I, Marjan ter Avest, in my capacity of director of MIND hereby confirm that MIND intends to support and contribute to the project PROactive TECHnology-supported prevention and MEntal health in adolescence (PROTECt ME project), as applied for by the main applicants, Loes Keijsers, Professor in Clinical Child and Family Studies at the Department of Psychology, Education and Child Studies, Erasmus School of Social and Behavioural Sciences at Erasmus University, Rotterdam and Manon Hillegers, Professor of Child and Adolescent Psychiatry, Erasmus MC, Rotterdam, The Netherlands.

A lot of our youth and youngsters suffer from various mental health problems, which have long lasting negative effects and cost our society an increasing budget, while the regular mental health care for youth is currently not effective, due to e.g. long waiting lists, monitoring procedures and administrative regulations. What is needed are structural changes in the approach to youth mental health care.

Therefore, we especially like about PROTECt Me, as has been described in the project summary “that the ground-breaking scientific insights into the root causes of mental health will immediately provide new ways to improve detection, decision making and intervention, new methods and approaches, and user-friendly and cost-effective solutions for youth and professionals. In the long-term, this results in proactive, personalized, accessible, and cost-effective prevention of mental health problems, based on a sustainable business proposition. The ultimate impact will be a decrease in adolescents entering specialized mental health care, shorter waiting lists and a significant reduction in societal costs of adolescent mental health problems. But most importantly; optimal well-being of all youth.”

This is of major importance to the beneficiaries and target groups of MIND. The PROTECt ME project therefore, supports the mission of MIND and with that also of MIND Young. The project has a perfect fit with the MIND Young platform that offers information, recognizable stories, news, tips and events about everything that has to do with your mind. By collaborating with MIND Young the project will have access to young people with whom co-creating sessions can be organized, as well as workshops. This will help with the dissemination of research results and online offered decision and intervention tools.

MIND Landelijk Platform Psychische Gezondheid | Stationsplein 125 | 3818 LE Amersfoort | (033) 303 24 00 | info@wijzijnmind.nl
www.mindplatform.nl | IBAN: NL44 FVLB 0227 222 598 | BTW: NL816859590B01 | KvK nummer: 30213449

MIND intents to provide an in-kind contribution towards the project costs within WP 3, in year 3-5. This contribution comprises of supporting co-creation sessions with youngsters regarding the design of mHealth applications, support the recruitment of participants to test new mHealth tools, contribution to the feedback sessions in order to improve the mHealth application, use of the MIND Young platform to disseminate results and youth participation experiences. The participating youngsters will receive travel costs and rewards for participating in the focus groups from the project budget. The contribution by MIND will represent a monetary value of € 10.000, in support of joint projects. This is in accordance with the further detailed project proposal and budget form.

Yours sincerely,

MSc. Marjan ter Avest
Director of MIND
March 8, 2022

Dean prof. V. Bekkers, Erasmus School of Social and Behavioural Sciences,
Erasmus University, Burg. Oudlaan 50, Rotterdam, The Netherlands

9 March 2022

LETTER OF INTENT
for the
PROTECt ME PROJECT

Dear prof. Hillegers and prof. Keijsers,

I, Lauwerens Metz , in my capacity of CEO at Innovattic hereby confirm that Innovattic BV intends to contribute to the PROactive TECHnology-supported prevention and MEntal health in adolescence (PROTECt ME project), as applied for by the main applicant, Loes Keijsers, Professor in Clinical Child and Family Studies at the Department of Psychology, Education and Child Studies. Erasmus School of Social and Behavioural Sciences at Erasmus University, Rotterdam.

The PROTECt ME project summary:

Societal problem. 13-25% of Dutch youth (age 12-25) suffer from depression or anxiety. Such mental health problems have long lasting negative effects, including educational dropout, worse physical health, and increased inequality. Mental health problems cost our society 2 billion euro per year. Adequate and timely identification and treatment are currently impossible, due to long waiting lists, time consuming monitoring procedures, and absence of widely and easily applicable personalized prevention strategies. Hence, systematic changes in our approach to youth mental health are sorely needed.

Scientific ambition: Optimizing well-being of Dutch youth aged 12-25 years by:

(WP 1) improving early identification of mental health problems, using cutting edge real-time measures, technology, and data modeling;

(WP 2) improving and implementing decision making of who needs which intervention and when, combining professional expertise with artificial intelligence;

(WP 3) creating and testing new technologically-driven interventions (e.g., eHealth and VR).

Our integrated approach involves stakeholders and end users in each step (WP 4); and builds a sustainable infrastructure and collaboration for convergence and impact (WP 5).

Output, outcome, impact. Our ground-breaking scientific insights into the root causes of mental health will immediately provide new ways to improve detection, decision making and intervention, new methods and approaches, and user-friendly and cost-effective solutions for youth and professionals. In the long-term, this results in proactive, personalized, accessible, and cost-effective prevention of mental health problems, based on a sustainable business proposition. The ultimate impact will be a decrease in adolescents entering specialized mental health care, shorter waiting lists and a significant reduction in societal costs of adolescent mental health problems. But most importantly; optimal well-being of all youth.

Fit to the convergence. Medical sciences, social science and humanities and technology converge to promote grand transitions in mental health care, for those who need us most; Our children, our future.

Innovattic BV intends to provide an in-kind contribution of consultancy for MARIO and/or for Generation R apps, module tracking teens, representing a monetary value of € 25000 and further detailed in the project proposal and budget form.

Yours sincerely,

Name: Lauwerens Metz

Position: CEO

Date: 09-03-2022

VO RAAD
Vereniging van scholen
in het voortgezet onderwijs

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E info@vo-raad.nl
www.vo-raad.nl

Dean prof. V. Bekkers
Erasmus School of Social and Behavioural Sciences
Erasmus University
Burg. Oudlaan 50
Rotterdam, The Netherlands

Subject: Letter of Intent for the PROTECt ME project

Utrecht, March 11 2022

Dear prof. Hillegers and prof. Keijsers,

I, Rens van den Boogaard, in my capacity of teamleader School & Environment at VO raad hereby confirm that VO-raad intends to contribute to the PROactive TECHnology-supported prevention and MEntal health in adolescence (PROTECt ME project), as applied for by the main applicant, Loes Keijsers, Professor in Clinical Child and Family Studies at the Department of Psychology, Education and Child Studies. Erasmus School of Social and Behavioural Sciences at Erasmus University, Rotterdam in collaboration with Prof. dr. Manon Hillegers, Department of Child and Adolescent Psychiatry/Psychology at the Erasmus MC Sophia, Rotterdam.

> The PROTECt ME project summary:
>
> Societal problem. 13-25% of Dutch youth (age 12-25) suffer from depression or anxiety. Such mental health problems have long lasting negative effects, including educational dropout, worse physical health, and increased inequality. Mental health problems cost our society 2 billion euro per year. Adequate and timely identification and treatment are currently impossible, due to long waiting lists, time consuming monitoring procedures, and absence of widely and easily applicable personalized prevention strategies. Hence, systematic changes in our approach to youth mental health are sorely needed.
>
> Scientific ambition: Optimizing well-being of Dutch youth aged 12-25 years by:
>
> (WP 1) improving early identification of mental health problems, using cutting edge real-time measures, technology, and data modeling;
>
> (WP 2) improving and implementing decision making of who needs which intervention and when, combining professional expertise with artificial intelligence;
>
> (WP 3) creating and testing new technologically-driven interventions (e.g., eHealth and VR).
>
> Our integrated approach involves stakeholders and end users in each step (WP 4); and builds a sustainable infrastructure and collaboration for convergence and impact (WP 5).

VO RAAD

Output, outcome, impact. Our ground-breaking scientific insights into the root causes of mental health will immediately provide new ways to improve detection, decision making and intervention, new methods and approaches, and user-friendly and cost-effective solutions for youth and professionals. In the long-term, this results in proactive, personalized, accessible, and cost-effective prevention of mental health problems, based on a sustainable business proposition. The ultimate impact will be a decrease in adolescents entering specialized mental health care, shorter waiting lists and a significant reduction in societal costs of adolescent mental health problems. But most importantly; optimal well-being of all youth.

Fit to the convergence. Medical sciences, social science and humanities and technology converge to promote grand transitions in mental health care, for those who need us most; Our children, our future.

VO-raad intends to provide an in-kind contribution to support crosslinks between schools and researchers in order to promote emotional wellbeing in students. This will include co-creation in research project, dissemination, supporting recruitment of mhealth study participating at all school levels (VMBO, MBO, HAVO, VWO), further detailed in the project proposal and budget form.

Yours sincerely,

Rens van den Boogaard
Teamleider School en Omgeving, VO-raad

JANTJE BETON
STOP NOOIT MET SPELEN!

Arthur van Schendelstraat 550
3511 MH Utrecht
T (030) 244 70 00
IBAN: NL 58 RABO 0160 8979 55
info@jantjebeton.nl
www.jantjebeton.nl
Beschermvrouwe Prinses Beatrix

Victor Bekkers
Erasmus University Rotterdam

betreft	plaats	datum
Statement of support	Utrecht	10th of March 2022

Convergence Flagship: PROactive TEChnology-supported prevention and MEntal health in adolescence (PROTECt ME)

Dear Victor Bekkers,

With this letter I declare on behalf of Jantje Beton our interest in and support for your research proposal PROactive TEChnology-supported prevention and MEntal health in adolescence (PROTECt ME).

As Jantje Beton we have been fully informed about your Convergence flagship application. We are very interested in supporting in this consortium.
Jantje Beton promotes wellbeing of children and youth through safe and independent access to outdoor play, sports and social gathering. Jantje Beton highly values the participation of youth as stakeholders in any decisions regarding their wellbeing and pays special attention to inclusion of all youth.
We believe that the overall aims of the research proposal and Jantje Beton to strive for optimal well-being of all youth align. We consider the active participation and empowerment of youth in research on this matter crucial for its success. Therefore, we support the cooperation between research and societal partners working with youth.
We state that we support the project and intend to share our expertise and network if possible.

Kind regards,

Pauline van der Loo
Manager Impact Jantje Beton (hoofd afdeling Impact Jantje Beton)
10th of March 2022

CBF ERKEND GOED DOEL

Convergence Flagship: PROactive TEChnology-supported prevention and MEntal health in adolescence (PROTECt ME)

Victor Bekkers

Erasmus University Rotterdam

Subject: Statement of support

Dear Victor Bekkers,

With this letter I declare on behalf of @ease our interest in and support for your research proposal PROactive TEChnology-supported prevention and MEntal health in adolescence (PROTECt ME).

As @ease we have been fully informed about your Convergence flagship application. We are very interested in supporting in this consortium.

At @ease, all young people between the ages of 12 and 25 can spontaneously walk in or chat if they need someone to talk to: without a waiting list, free and anonymously. Trained young volunteers and professionals are ready to support them. Simply by being there and listening! And if necessary, we help find the right care outside @ease.

We believe that the overall aims of the research proposal and @ease to support the well-bein of all youth align. We consider the active participation if youth and practitioners and empowerment of youth in research on this matter crucial for its success. Therefore, we support the cooperation between research and societal partners working with youth.

We state that we intend to actively participate as advisors of the project as to share our knowledge, expertise, and networks.

11-3-2022

Ruth Rietveld

Management @ease Rotterdam

R Rietveld

Datum 10 maart 2022
Telefoon
E-mail onz@tudelft.nl
Onderwerp Letter of support Convergence Flagship
Reference TNW 22.202

Delft University of Technology

Science Education & Communication
Faculty of Applied Sciences

Bezoek

Lorentzweg 1
2628 CJ Delft
Correspondentie
Postbus 5046
2600 GA Delft

www.sec.tudelft.nl

Convergence Flagship:
PROactive TEChnology-supported prevention and MEntal health in adolescence (PROTECt ME)
Victor Bekkers
Erasmus University Rotterdam

Dear Victor Bekkers,

With this letter I declare on behalf of Bètasteunpunt Zuid-Holland our interest in and support for your research proposal PROactive TEChnology-supported prevention and MEntal health in adolescence (PROTECt ME).

As Bètasteunpunt Zuid-Holland we have been informed about your Convergence flagship application. We are very interested in supporting this consortium.
Bètasteunpunt Zuid-Holland promotes professionalization of teachers in secondary education in the beta and technology domain, including 'Ontwerpend en onderzoekend leren'.
We regularly collaborate with schools to disseminate our knowledge and test our methodologies at schools. As such we have a very good and enthusiastic network in secondary education nationally.

We believe that the overall aims of the research proposal to strive for optimal inclusion of children and youth in the development of e-health interventions for their well-being aligns well with the aims of the Bètasteunpunt Zuid-Holland to teach youth through executing design projects. Therefore, we support the cooperation between our teacher professionalisation efforts and societal partners working with youth.
We state that we intend to actively participate as advisors of the project as to share our knowledge, expertise, and networks of secondary schools, as it contributes to teacher professionalization.

Yours sincerely,

Prof.dr.ir. P.M. Herder
Dean of the Faculty of Applied Sciences
Bètasteunpunt Zuid-Holland

Mathieu Gielen - IO   Monday, March 14, 2022 at 11:24:59 Central European Standard Time

Onderwerp: Wetenschapsknooppunt joins the Betasteunpunt in supporting PROTECt-ME
Datum: vrijdag 11 maart 2022 om 14:03:33 Midden-Europese standaardtijd
Van: Remke Klapwijk - TNW
Aan: Mathieu Gielen - IO

Dear Mathieu Gielen,

I am writing you to inform you that our dean, Prof.dr.ir. Paulien Herder, has agreed that the Wetenschapsknooppunt Delft will join the Betasteunpunt Zuid-Holland in its support for your flagship proposal "PROactive TEChnology-supported prevention and MEntal health in adolescence" (PROTECt-ME). With both our parties, you now have the support of the Science Education and Communication Department regarding its knowledge and network towards both Primary Education and Secondary Education.

We believe that co-design with children will benefit innovations regarding children's daily life while also benefiting children directly in acquiring important skills. We subscribe to the statement of support written by Betasteunpunt in their letter and we will be glad to support you in achieving participation of children and youth in innovations that address their mental health care.

Please accept this message as a letter of support and relay it to Victor Bekkers of Erasmus University.

Kind regards,

On behalf of Prof. Herder,

Remke Klapwijk
Principal researcher at Wetenschapsknooppunt TU Delft, SEC
Friday 11 March 2022

Page 1 of 1

III Letter of support from department head of the main lead

Erasmus School of
Social and
Behavioural Sciences

To whom it may concern

Date
11 March 2022

Subject
Support letter prof. dr. L. Keijsers

Our reference
ESSB/VB/21453/nbo

Your reference

Page
1/1

Appendix

Department
Dean's Office

Visiting address
Erasmus University Rotterdam
Burgemeester Oudlaan 50
Mandeville
T15-19

Postal address
PO Box 1738
3000 DR Rotterdam
The Netherlands

T +31 10 408 2064
E office.dean@essb.eur.nl
W www.eur.nl/essb/English

To whom it may concern

I, professor Victor Bekkers, in my capacity as dean of the Erasmus School of Social and Behavioural Sciences (ESSB), hereby confirm that the faculty intends to fully support the main applicant, Prof. Dr. Loes Keijsers, full professor at Erasmus University Rotterdam, if the proposal "PROactive TEChnology-supported prevention and MEntal health in adolescence", is funded by the Convergence Health & Technology Flagship program.

The PROTECT ME project fits perfectly with the research agenda of the faculty's Department of Psychology, Education & Child Studies and Erasmus University's goal of creating positive societal impact. The ESSB looks forward to joining forces with the Erasmus Medical Centre and the TU Delft to improve young people's mental health by supporting this exciting new research program at the interface of social sciences, the humanities, medicine, and technology.

The ESSB will commit the following support if the project is funded:

- Hosting the main applicant for the duration of the project
- Providing research support for the main applicant and team members during the execution of the project, including the use of research infrastructure, project management support, advice and support for research funding acquisition, privacy assessments and data management support.

Yours sincerely,

Prof. dr. V.J.J.M. Bekkers
Dean of the Erasmus School of Social and Behavioural Sciences,

Erasmus University Rotterdam

IV References

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